

**SYSTEM DEVELOPMENT FOR HUMAN-IN-THE-LOOP MACHINE
LEARNING EXPERIMENTS TO INCREASE VIDEO ACCESSIBILITY FOR
VISUALLY IMPAIRED AND BLIND USERS**

A CULMINATING EXPERIENCE REPORT

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Abstract

This project focuses on improving video accessibility for blind and visually impaired users by providing the volunteering describers AI assistance via a web interface to produce high-quality audio narrations for YouTube Videos. Video accessibility is crucial for blind and visually impaired individuals for education, employment, and entertainment purposes. However, asking professionals to create video descriptions is a costly and time-consuming affair. The alternative is to ask volunteers to create these descriptions. The downside of asking volunteers to manually come up with descriptions is that the quality varies depending upon each person's aptitude for the task and it also tends to be an intimidating challenge for novice describers. This project proposes a Human-in-the-Loop Machine Learning (HILML) approach to video description by automating video text generation and scene segmentation and allowing humans to edit the output. The HILML approach facilitates human-machine collaboration to produce high quality video descriptions while keeping a low barrier to entry for volunteer describers. This AI-infused system was proven to be significantly faster and easier to use for novice describers as per a user study conducted by the project team. The quality of the video descriptions and understanding of the topic created by the HILML system compared to manually generated descriptions were rated as being significantly higher by blind and visually impaired users.

Keywords: Video Accessibility; Video Description; YouDescribe; Visually Impaired Users; Artificial Intelligence; Human-in-the-Loop;

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1. Introduction

Audio description is a powerful tool for allowing people with disabilities to fully participate in the educational, social, cultural, economic and political lives of their communities. The benefits of audio description for videos are wide-reaching. They are a classic example of the universal design philosophy, which states that “what is essential for some, is almost always helpful for all.” Video accessibility for blind or visually impaired users becomes crucial as online videos on websites such as YouTube come to play an essential role in our daily lives. However, the vast majority of video material on the Internet is currently not accessible to these millions of visually impaired people who would significantly benefit from improved access to videos for education, employment, social, and entertainment purposes, thus creating an increasing digital divide between blind and sighted users [1].

Described media has recently become available in the mainstream, and less is known about its impact beyond people who are blind. Another potential beneficiary of described content could be students with autism. The theory is that they could benefit from the more explicit explanations (especially with regard to emotions) in the audio descriptions. Also, as more people consume their video content in places where it is not always possible to look at a screen (thanks in part to the rise of mobile technology) the usefulness of described media as a universal design feature will become more evident. Already, many of us are becoming used to interacting with screen-less devices in the form of smart speakers such as the Amazon Echo or the Google Home.

In this project, we advocate the use of Human-in-the-Loop Machine Learning (HILML) to make the process easier for novice describers. This would lead to increased productivity and would eventually result in the improvement in the quantity and quality of accessible videos for blind and visually impaired users. A video description interface was built which relied on automated scene segmentation, caption generation from video-frames, and Optical Character Recognition (OCR) using machine learning. The volunteer(describer) is then able to use the interface to edit the machine-generated text, thereby facilitating human-machine collaboration to produce high quality video descriptions while ensuring that the barrier to entry for volunteer describers is kept low.

This interface was evaluated on novice describers and compared to a control condition with no machine learning assistance (User Study - Phase 1). Then, blind and visually impaired users were asked to watch specific videos with synchronized audio descriptions produced by the two conditions (User Study - Phase 2).

Thus, the contributions of this project are as follows:

1. Design and develop a Human-in-the-Loop Machine Learning (HILML) system to generate video descriptions for visually impaired and blind users.
2. Demonstrate that the HILML system is faster and easier to use for novice video describers than a human only control condition.
3. Demonstrate that the quality of the video descriptions and topic understanding is significantly higher for the HILML system for blind and visually impaired raters compared to a human only control condition.

2. Related Work

In this section we discuss existing methods and work in the field of HCI, video accessibility and AI aimed at facilitating an improved experience for the blind and visually impaired people in their everyday lives.

2.1 Similar Tools - LiveDescribe

LiveDescribe was the first known computer-assisted video description [2]. It was designed for real-time descriptions of live television broadcasts. The two automated features of LiveDescribe were:

- It could automatically detect silent periods where a describer could choose to add a description,
- LiveDescribe would allow descriptions to go beyond the silent periods by automatically pausing the video in order to accommodate extended descriptions.

Researchers made amateur describers with little or no previous description experience use the software Live Describe to describe a single episode of a 20-minute comedy show. Seventy-five reviewers who were blind, had low vision, or were sighted then rated the descriptions using a number of criteria, including overall quality and entertainment value [3]. It was noted that the listeners' preferences for description seem to be based on various characteristics of describers, such as the describers' vernacular and tone of voice and the length and timing of the descriptions.

The primary difference between LiveDescribe and our work is:

- LiveDescribe focuses on real-time descriptions of live TV broadcasts whereas our work is not focused on describing Live events. The AI components require some time to extract labels, objects, captions, scene segments etc. from the video
- Our project focuses on using AI detecting scenes with specific start times and scene times. LiveDescribe relied on measuring the audio output to decide whether it is a silent period where descriptions could be accommodated.
- Feedback from describers was positive in both cases but high cognitive workload was reported by the participants of the user study conducted for LiveDescribe

whereas the users of our prototype reported reduced cognitive workload as compared to a human-only system.

2.2 Human-in-the-Loop Machine Learning

This is a branch of artificial intelligence that leverages both human and machine intelligence to create machine learning models. The human-in-the-loop approach combines the best of human intelligence with the best of machine intelligence. Machines are great at making smart decisions from vast datasets, whereas people are much better at making decisions with less information.

HILML has been used in fields such as assistive robotics [4], data analytics [5], and audio labeling [6]. Researchers have used HILML to create a semi-automatic method to segment foreground moving objects in surveillance videos [7]. They combined human outlining of a small number of moving objects with a convolutional neural network to reach similar levels of accuracy as a human with less manual work than human-only analysis.

In this project, both human and machine intelligence were leveraged to create a HILML system to aid sighted humans create video descriptions for blind or visually impaired users.

3. Motivation

The primary difficulty faced while adding video accessibility features is that developing professional quality video descriptions is extremely time consuming and mentally exhausting. Adding descriptions to videos necessitates a certain degree of expertise to sufficiently describe the scenes, actions and nuances within the video. Tasks such as being able to differentiate between important and unimportant details, synchronizing the resultant descriptions with the actual video tend to be difficult and put a great amount of cognitive workload on the volunteering describer. Hence, video description tends to be a challenging task for inexperienced volunteers who may be limited by their own aptitude, vocabulary, or lack of training.

‘Be My Eyes’ [8] is an existing application for mobile platforms that connects blind and visually impaired people with over two million sighted volunteers through video calling. It is one of the biggest online communities for blind and visually impaired people. In comparison, the ‘YouDescribe’ [9] is a free platform having roughly 2500 sighted volunteers that have described more than 4000 YouTube videos till date at the request of their user base of blind users. This large discrepancy in the number of volunteers along with informal feedback from volunteers indicates that volunteers are willing to help blind and visually impaired users, but describing videos is not something that are able to or willing to currently do easily. According to volunteers, difficulty of the task and the skills needed such as language and/or writing skills, combined with the public exposure of their work, were deterrents to them volunteering for video description. Hence, there is a great variance in the quality of video descriptions on the YouDescribe platform depending on each volunteer’s aptitude for the task.

There is an acute need for a system that automates certain parts of the process of description-generation. This system should be suitable for volunteers(describers) of all skill levels and experience. Volunteers face a significant amount of fatigue and cognitive load while describing videos. Moreover, description as a task also tends to be time consuming. The system should be able to solve the above-mentioned issues satisfactorily so as to encourage more people to volunteer for description tasks - thereby improving accessibility for the blind and visually impaired.

4. Application Design

In this section we discuss in detail the system architecture and then briefly review the functionalities of each step in the system to assist human volunteers in producing video descriptions using a HILML (Human in the Loop Machine Learning) approach. The developed system relies on the usage of publicly available APIs such as Microsoft Azure Cognitive Services [10], Google Cloud Services [11], YouTube IFrame API [12] and the IBM Max Scene Classifier (Places365 CNN) [13] for computer vision and language. These APIs provide capabilities for scene detection, generation of descriptions(captions) for individual frames, face and emotion detection, Optical Character Recognition, Text-to-Speech conversion and Speech Synthesis. The framework utilizes the above-mentioned tools to effectively extract information from video clippings and generate appropriate descriptions. The system is adept at detecting scene changes, detecting people and objects within the scenes and coming up with textual descriptions for the occurrences within each scene.

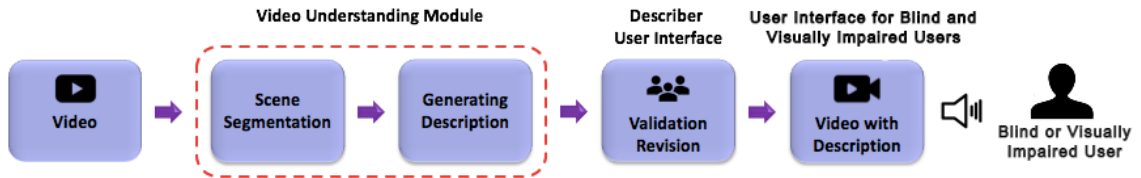


Figure 1. The workflow of the HILML system

The following describes the workflow (Figure 1) of the developed system:

1. **Video Input:** Videos for which descriptions have been requested by blind/visually impaired users are forwarded to the system for further processing.
2. **Video Understanding Module:** This module is responsible for extraction of video metadata by applying various ML APIs and pipelining this output to another set of publicly available APIs to get a final description. It has 2 parts:
 - a. **Scene Segmentation and Key Frame Extraction:** The video is segmented into a sequence of scenes of varying time spans, and key

frames are sampled to maintain the appropriate granularity of the scene for generating the description.

- b. **Description Generation:** Sequences of key frames are processed by the model to generate descriptions that best explain the scene in the video. The description covers any text in the key frames, people with ID (to handle reappearances), gender, emotion, hair color, age, objects, and environment. People are recognizable if they are known celebrities.

3. **Description Revision & Validation via UI:** Through the describer(volunteer) interface, volunteers (sighted users) can view video scene segments, their associated descriptions, and can make appropriate changes/additions to the machine generated text by editing the contents of the pre-populated text fields. This step would assist in improving the quality of narrations that are eventually presented to blind and visually impaired users. The interface along with the pre-filled text fields assist in reducing the cognitive load on the volunteers. Hence, it provides a well-structured template whose contents can be modified as deemed necessary by the volunteers which makes it greatly superior to free-form descriptions that volunteers create from scratch.
4. **Video Playback with Synchronized Description Audio:** The video is played to the visually impaired or blind user with audio descriptions synchronized to the original video. Audio is generated using Google Cloud Service's text-to-speech API [14]. Synthesized video descriptions have been shown to be acceptable to blind and visually impaired users and are used by professional video description production companies.

The above can be briefly summarized as:

- The system takes a video file as input for further processing
 - Scene segments within the video are detected using ML APIs
 - Captions (Descriptions) are generated from key-frames within each scene.
- Also, OCR detection is used to detect on-screen text.

- The textual descriptions are synthesized to speech (audio files). The generated audio descriptions are then synchronized with the video as per the detected scene segments and played for the blind/visually-impaired user.

5. Implementation

5.1. Tools and Libraries used

- Cloud service used for project deployment: Amazon AWS EC2 [15]
 - Operating System: Ubuntu Server 18.04 LTS 64-bit (x86)
 - EC2 Instance details:
 - Type: t2.micro
 - Number of virtual CPUs: 1 vCPU @ 2.5GHz (Intel Xeon Family)
 - Memory (RAM): 1GB
 - Public IP address: 18.221.192.73
 - Public DNS: ec2-18-221-192-73.us-east-2.compute.amazonaws.com
 - EC2 console instance tag: sfsu_youdescribe
 - Availability Zone: us-east-2c
 - VM Instance Storage: 30GB
 - Mounted EBS instance Storage: 50GB
 - Security Group:
 - Name: Youdescribe-Security-Group
 - Group ID: sg-0b3c6ed6b3c2f3bae
 - **Table 1: AWS Outbound Traffic Rules:**

Type	Protocol	Port Range	Destination	Notes
All Traffic	All	0 - 65535	0.0.0.0/0, ::/0	VM can make outbound requests to any IP at any port

- **Table 2: AWS Inbound Traffic Rules:**

Type	Protocol	Port Range	Source	Notes
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SSH	TCP	22	0.0.0.0/0, ::/0	Enable SSH connections from any IP address
HTTP	TCP	80	0.0.0.0/0, ::/0	Accept standard incoming HTTP and HTTPS requests from any IP address on the standardized ports 80 and 443
HTTPS	TCP	443	0.0.0.0/0, ::/0	
Custom	TCP	5000	0.0.0.0/0, ::/0	Accept incoming HTTP requests on Ports 5000, 5001, 5002 from any IP address
Custom	TCP	5001	0.0.0.0/0, ::/0	
Custom	TCP	5002	0.0.0.0/0, ::/0	

- Runtime environment: Node JS [16]
 - Node Version: v10.15.3
 - NPM Version: v6.4.1
- Web Development Framework (Backend): Express JS [17]
 - Express JS Version: v4.16.0
- Database: PostgreSQL [18] RDBMS (Via Amazon RDS) [19]
 - Type: db.t2.micro
 - Allocated Storage: 20GB
 - Storage Type: General Purpose (SSD)
 - Database Port: 5432
 - PostgreSQL Version: v11.1-R1
- Web Frontend technologies/major libraries:

- HTML5, CSS
- JavaScript (ECMAScript 6 Standard)
- JQuery 3.4.1, Bootstrap 4.3.1, YouTube Iframe API [12]
- AI Components:
 - Microsoft Azure Cognitive Services [10]:
 - Video Indexer API (Video Intelligence API v1.5.1) [20]
 - Computer Vision API v2.0 [21]
 - Face API v2.0 [22]
 - Text-To-Speech API v1.3.0 [14]
 - IBM Max Scene Classifier v1.1.0 [13]

5.2 Database Schema

The PostgreSQL database consists of 12 tables. Brief descriptions of the core use cases for each table/entity:

- **Video:** Used to store data about the video such as title, duration, YouTube ID, etc. Deleting a row in the video table has a cascading effect on the blocks, scenes, shots, keyframes associated with it.

Video	
id	integer
youtube_id (PK)	varchar
indexer_video_id (Unique)	varchar
title	varchar
duration	real
summarised_insights_json	varchar
insights_json	varchar

- **Block:** Each Video consists of multiple blocks (each block has a specific start and end time) - This table is currently unused in its current form due to inconsistencies in data returned by the video indexer API which may get fixed in the future.

Block	
id	integer
block_id (PK)	varchar
video_id (FK)	varchar
block_num	integer
start_time	real
end_time	real
original_description	varchar

- **Block_Modification:** This table is supposed to be used to track modifications made to AI generated block descriptions by the volunteers. It is not being used in the application currently but could be utilized in the future.

Block_Modification	
id	integer
block_modification_id (PK)	varchar
block_id	varchar
user_id (FK)	varchar
modified_description	varchar

- **Scene:** Each Video consists of multiple scenes (each scene has a specific start and end time). Once the Video Indexer API matures, scenes are expected to be subsets of blocks.

Scene	
id	integer
scene_id (PK)	varchar
block_id (FK)	varchar
scene_num	integer
start_time	real
end_time	real
original_description	varchar
original_ocr	varchar

- **Scene_Modification:** This table is used to track modifications made to AI generated scene descriptions by the volunteers.

Scene_Modification	
id	integer
scene_modification_id (PK)	varchar
scene_id (FK)	varchar
user_id (FK)	varchar
modified_start_time	real
modified_end_time	real
modified_description	varchar
modified_ocr	varchar

- **Shot:** Each scene in the video is made up of multiple shots (each shot has a specific start and end time).

Shot	
id	integer
shot_id (PK)	varchar
scene_id (FK)	varchar
shot_num	varchar
start_time	real
end_time	real
original_ocr	varchar
original_description	varchar

- **Keyframe:** Key-Frames(images) may be extracted from shots for further analysis. This table stores details about the image such as its storage location, timestamp along with an AI generated description of the image among other fields.

Keyframe	
id	integer
keyframe_id (PK)	varchar
shot_id (FK)	varchar
keyframe_num	varchar
start_time	real
end_time	varchar
thumbnail_id	varchar
thumbnail_url	varchar
caption	varchar
caption_confidence	real

- **Keyframe_Object:** This table stores details of all objects detected in the keyframe such as name, confidence score and position coordinates.

Keyframe_Object	
id	integer
keyframe_object_id (PK)	varchar
keyframe_id (FK)	varchar
object_name	varchar
confidence	real
x_coordinate	real
y_coordinate	real
width	real
height	real

- **Keyframe_Tag:** This table stores details of all objects detected in the keyframe such as name, confidence score. Unlike objects, the VideoIndexer API doesn't return position coordinates for tags.

Keyframe_Tag	
id	integer
keyframe_tag_id (PK)	varchar
keyframe_id (FK)	varchar
tag_name	varchar
confidence	real

- **Keyframe_Environment:** This table stores the IDs of the top 5 detected environments for each keyframe.

Keyframe_Environment	
id	integer
keyframe_environment_id (PK)	varchar
keyframe_id (FK)	varchar
environment_id (FK)	integer
probability	real

- **Environment:** This is a pre-populated reference table that contains 365 possible environments that can be detected by the Places 365 CNN each environment type has a unique ID.

Environment	
id	integer
environment_id (PK)	integer
category_name	varchar
level1_in	integer
level1_out_natural	integer
level1_out_manmade	integer
level2_in_shopping_and_dining	integer
level2_in_workplace	integer
level2_in_home_or_hotel	integer
level2_in_transportation	integer
level2_in_sports_and_leisure	integer
level2_in_cultural	integer
level2_out_n_water_ice_snow	integer
level2_out_n_mountains_hills_deserts_sky	integer

level2_out_n_forest_field_jungle	integer
level2_out_n_manmade_elements	integer
level2_out_m_transportaion	integer
level2_out_m_cultural_or_historical	integer
level2_out_m_sportsfield_park_leisure	integer
level2_out_m_industrial_and_construction	integer
level2_out_m_houses_cabins_gardens_farms	integer
level2_out_m_commercial_shops_cities_towns	integer
level2_out_m_commercial_shops_cities_towns	integer

- **User:** This table stores details for all users who have volunteered to assist in the video description process.

Us_User	
id	integer
uuid (PK)	varchar
name	varchar

- **Us_User_Video:** This table maps user IDs to the videos that have been assigned to each describer involved in Phase 1 of the user study. It also indicates whether an AI generated description has to be shown or not.

Us_User_Video	
id	integer
user_id (FK)	varchar
video_id	varchar
ai_support	boolean

- **Us_User_Blind:** A table to store data of visually impaired/blind users.

Us_User_Blind	
id	integer
uuid (FK)	varchar
name	varchar
submitted	boolean
access_preference	varchar
vid_desc_frequency	varchar
email	varchar
access_pref_other	varchar

- **Us_Helper_Video:** Table that maps visually impaired/blind users to specific videos that they are supposed to evaluate.

Us_Helper_Video	
id	integer
volunteer_id (FK)	varchar
video_id (FK)	varchar
ai_support	boolean
volunteer_name	varchar
rating	integer
comment	varchar
vi_user_id (FK)	varchar
vi_user_name	varchar
video_understanding	integer

5.3. Video Understanding Module

The Video understanding module is responsible for scene segmentation and description generation for the input video. This is a multi-stage process and involves the usage of several Microsoft [10][20][21][22] and IBM [13] machine-learning APIs in conjunction with video and image manipulation tools. It involves the execution of a series of time-consuming operations in a specific sequence to facilitate the extraction of metadata and descriptions from the video. This data is then aggregated, cleaned (removal of duplicate OCR text detections) and saved into the PostgreSQL database. In its current state, the project doesn't utilize certain parts of the extracted metadata due to changed requirements but still stores them in the database in case it is needed to build a new feature in future versions of the project.

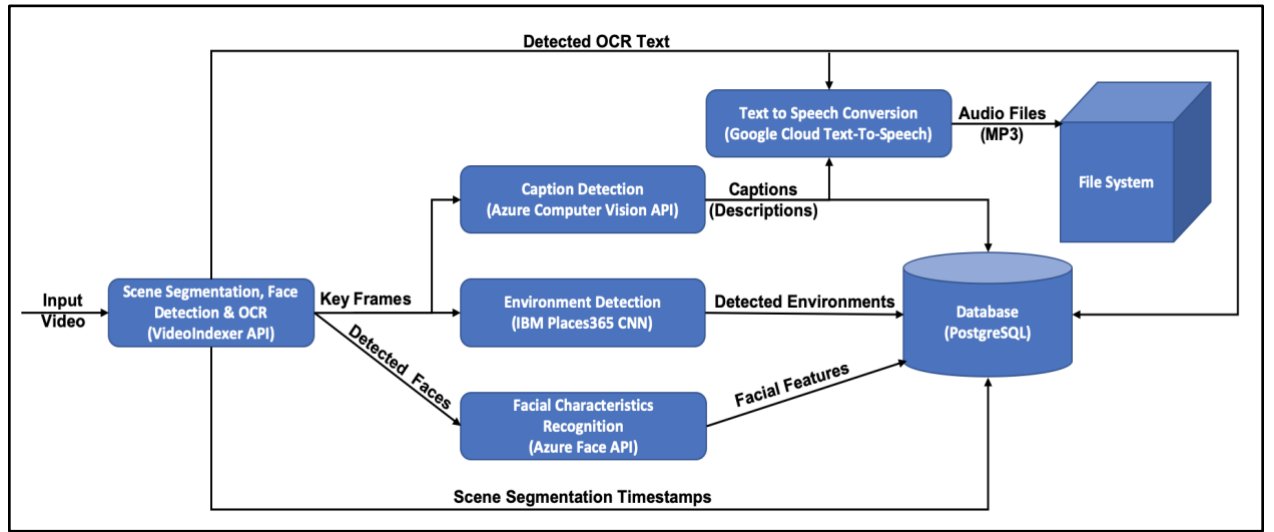


Figure 2: Video Understanding Module's Architecture

5.3.1. Scene Segmentation, Optical Character Recognition (OCR), Face detection

The Microsoft Azure Video Indexer API [20] is extensively used for the scene-segmentation process. The API supports segmenting videos into temporal units based on structural and semantic properties. The primary goal of this step is to recognize timestamps at which there is a change of scene within the input video.

The API returns metadata called ‘insights’ to indicate the timestamps for shots, scenes and key-frames and the corresponding OCR text detected from each shot. A brief description of each type:

- **Shot:** A continuous sequence of frames acquired by a single camera is called a shot, which is the basic building blocks of a movie. The Video Indexer API [20] also returns the text detected at various timestamps in the shot.
- **Scene:** It consists of one or more consecutive shots, which are semantically correlated and share the same physical settings or presents a continuous action performed by the actors. Thus, it’s often easier to recall a scene rather than a shot, because it covers a whole event or sub-story in a movie.
- **Keyframe:** Each shot may consist of one or more keyframes. The Video Indexer API [20] selects the frame(s) that best represent each shot. Keyframes are the representative frames selected from the entire video based on aesthetic properties (for example, contrast and stableness). The API also returns image files corresponding to faces detected in the keyframe.
- **Faces:** The Faces API [22] is able to recognize unique faces and returns thumbnail images of the face as well as the time slots during the video indicating each occurrence of the person. The API is also able to recognize celebrities/famous people and returns their names. If it is unable to recognize the face, a generic name such as “Unknown #1”, “Unknown #2”, etc. is assigned.

In its current form, the API JSON is downloaded manually from the [videoindexer.ai](https://api-portal.videoindexer.ai/) website. However, in future versions the process of uploading a video and downloading its insights can be automated as the official website(<https://api-portal.videoindexer.ai/>) contains specific API endpoints which can be used for this purpose.

Hierarchy of important insights nested within the JSON data:

```
{
  "videos": [
    {
      "insights": {
        "ocr": [..array..],
        "blocks": [..array..],
        "scenes": [..array..],
        "shots": [
          {
            "keyframes": [..array..]
            ...
          },
          ...
        ],
        ...
      }
    }
  ],
  ...
}
```

Example of the JSON returned from the API after giving it a video file (MP4) as input

```
{
  "id": "97eba92178",
  "name": "Shoplifting Scene - Ocean's 8",
  "durationInSeconds": 143,
  ...
  "summarizedInsights": {
    "thumbnailId": "5c334e6b-2e72-4205-a6e5-823b0e869fa6",
    "faces": [..array..],
    "keywords": [..array..],
    "sentiments": [..array..],
    "emotions": [..array..],
    "labels": [..array..],
    "brands": [..array..],
    "namedLocations": [..array..],
    "namedPeople": [..array..],
    "statistics": {..object..},
    "topics": [..array..]
  },
  ...
},
"videos": [
  {
    "id": "97eba92178",
    "insights": {..object..},
    "language": "en-US",
    "isAdult": false,
    "viewToken": "..token string here.."
    ...
  }
],
"videosRanges": [..array..]
}
```

It is essential to note that the “summarisedInsights” object contains arrays called “faces” and “emotions” which are currently ignored by the system that has been built. The reason being that this data is not granular enough. Instead, data corresponding to “faces” is taken from within the “insights” object contained within the first object of the “videos” array. Moreover, a separate API is used for detecting the emotions of the above mentioned “faces”.

Example of data within the “insights” JSON object

```
{
  "duration": "0:02:23.296",
  "language": "en-US",
  "transcript": [..array..],
  "ocr": [..array..],
  "keywords": [..array..],
  "topics": [..array..],
  "faces": [..array..],
  "labels": [..array..],
  "scenes": [..array..],
  "shots": [..array..],
  "sentiments": [..array..],
  "emotions": [..array..],
  "visualContentModeration": [..array..],
  "blocks": [..array..],
  "framePatterns": [..array..],
  "speakers": [..array..],
  "textualContentModeration": {..object..},
  "statistics": {..object..},
  ...
  ...
  ...
}
```

Expanding the “insights” JSON object, we get the most important part of the meta data which contains JSON arrays representing all blocks, scenes, shots. OCR-text and faces detected within the entire video

Scenes JSON: Sample of data seen after expanding the “scenes” JSON Array nested within the “insights” JSON Array.

```
"scenes": [
  ...
  ...
  ...
  {..object..},
  {..object..},
  {
    "id": 3,
    "instances": [
      {
        "start": "0:00:49.149",
        "end": "0:00:59.66"
      }
    ]
  }
]
```

```

    ]
  },
  {
    "id": 4,
    "instances": [
      {
        "start": "0:00:59.66",
        "end": "0:01:21.148"
      }
    ]
  },
  {..object..},
  {..object..},
  ...
  ...
]

```

Shots JSON: Sample of data seen after expanding the “shots” JSON Array nested within the “insights” JSON Array.

```

"shots": [
  ...
  ...
  ...
  {..object..},
  {..object..},
  {
    "id": 3,
    "tags": [
      "Indoor"
    ],
    "keyFrames": [..array..],
    "instances": [
      {
        "start": "0:00:20.787",
        "end": "0:00:23.79"
      }
    ]
  },
  {
    "id": 4,
    "tags": [
      "Indoor",
      "Wide",
      "LeftFace"
    ],
    "keyFrames": [..array..],
    "instances": [
      {
        "start": "0:00:23.79",
        "end": "0:00:28.862"
      }
    ]
  },
  {..object..},
  {..object..},
  ...
  ...
  ...
]

```

It should be noted that “shots” are a separate array from “scenes” as is not nested within “scenes”. Hence, it is the task of the JSON parsing code to determine the parent scene of each shot.

Keyframes JSON: Sample of data seen after expanding the “keyframes” JSON Array nested within each object in the “shots” JSON Array.

```
"keyframes": [  
  ...  
  ...  
  ...  
  { ..object.. },  
  { ..object.. },  
  {  
    "id": 7,  
    "instances": [  
      {  
        "thumbnailId": "739b74b4-a834-483c-84df-3f4f654cfd54",  
        "start": "0:00:21.588",  
        "end": "0:00:21.621"  
      }  
    ]  
  },  
  { ..object.. },  
  { ..object.. },  
  ...  
  ...  
  ...  
]
```

It is important to note that “instances” is an array of objects and not an individual object. This takes care of the edge-case where the same frame may be reused within different segments of the video. However, since reuse of frames is an unlikely case - in a vast majority of cases, the “instances” array contains just one object.

Faces: Sample of data seen after expanding the “faces” JSON Array nested within each object in the “insights” JSON Array.

```
"Faces": [  
  ...  
  ...  
  ...  
  { ..object.. },  
  { ..object.. },  
  {  
    "id": 1002,  
    "instances": [  
      ...  
      ...  
      ...  
      { ..object.. },  
      { ..object.. },  
      {  
        "thumbnailsIds": [  

```

```

        "6955a59f-9d78-41e4-b174-c6e142c53198"
    ],
    "start": "0:00:37.838",
    "end": "0:00:40.507"
  },
  {..object..},
  {..object..},
  ...
  ...
  ...
],
"name": "Sandra Bullock",
"confidence": 0.9612,
"description": "Sandra Annette Bullock is an American actress, producer ...",
"thumbnailId": "1c455209-0d58-4d16-9687-f673023b3962",
"referenceId": "02e07452-35e9-a82c-f32e-db9dadf6a616",
"referenceType": "Bing",
"title": "American Actress",
"imageUrl":
"https://www.bing.com/th?id=AMMS_f2e59c9fd2fed50c4d1fd78a261012c7&c=12&rs=1&qlt=80&cdv=1&pid=16.2",
"thumbnails": [..array..]
},
{..object..},
{..object..},
...
...
...
]

```

This JSON Array contains multiple “face” json objects which contain instances at which a particular person’s face was visible in the video and also provides corresponding thumbnail images. These thumbnail images can be used at a later stage to determine emotions of the character within a particular segment of the video using another API.

OCR: Sample of data seen after expanding the “ocr” JSON Array nested within each object in the “insights” JSON Array.

```

"ocr": [
  ...
  {..object..},
  {
    "id": 3,
    "text": "With words and narration by",
    "confidence": 0.9106,
    "left": 368,
    "top": 293,
    "width": 545,
    "height": 45,
    "language": "en-US",
    "instances": [
      {
        "start": "0:00:25.92",
        "end": "0:00:26.4",
        ...
      }
    ]
  },
  {..object..},
  ...
]

```


The object contains information that indicates the placement of text on the screen (left, top, width, height) - These XY coordinates rely on considering the top-left corner of the screen as having coordinates (0,0). This OCR data cannot be blindly parsed and used since successive objects may have the same text (As the same text may remain on the screen for a long length of time) - Hence it is essential to intelligently parse this data and ensure that there are no duplicates. The current system ignores all duplicates (i.e., having the exact same text and position on screen) within a time range of 3 seconds. The process of elimination of duplicates is not perfect and is further made difficult by fade-in/fade-out effects of text within videos as these effects result in detection of incorrect and incomplete OCR.

**Code for cleaning up OCR data within each shot (Contained in:
/backend/video.js):**

```
function cleanUpOcrDataWithinEachShot(ocrArr) {
  let ocrMap = new Map();
  for(let i=0; i<ocrArr.length; i++) {
    let currOcrObj = ocrArr[i];
    //currOcrObjKey is generated by appending certain important attribute values to form a unique
    string - useful for finding OCR detections with similar data
    let currOcrObjKey = currOcrObj.shot_id + '_' + currOcrObj.text + '_' + currOcrObj.left + '_' +
    currOcrObj.top + '_' + currOcrObj.width + '_' + currOcrObj.height;
    if(ocrMap.has(currOcrObjKey)) {
      let mOcrObj = ocrMap.get(currOcrObjKey);
      //Combine OCR shots properties if time difference between similar OCR detections is less
      than 3 seconds (the text, position on screen and size are already same)
      if(currOcrObj.start_time - mOcrObj.end_time < 3) {
        mOcrObj.end_time = currOcrObj.end_time;
      } else {
        currOcrObjKey = currOcrObjKey + '_' + i;
        ocrMap.set(currOcrObjKey, currOcrObj)
      }
    } else {
      ocrMap.set(currOcrObjKey, currOcrObj);
    }
  }
  let outputArr = [];
  let ocrMapKeys = ocrMap.keys();
  for(const mapKey of ocrMapKeys) {
    let ocrObj = ocrMap.get(mapKey);
    outputArr.push(ocrObj);
  }
  return outputArr;
}
```

Code to parse and insert all JSON data into the database (Contained in: /backend/video.js)

```
async function addVideoDataToDb(inputJson) {
  try {
    // //Parse and insert Video
    let video = Video.parseVideoFromIndexerJson(inputJson);
    console.log({VIDEO: video});
    await Video.insertIntoVideoTable(video);

    //Parse and insert Blocks
    let blockJsonArr = video.insights_json.blocks;
    let blockArr = Block.parseBlockArr(blockJsonArr, video.youtube_id);
    await Block.insertIntoBlockTable(blockArr);
    let blockArr = Block.parseBlockArr(blockJsonArr, video.youtube_id);
    blockArr = modifyLastElementsOfBlockArray(blockArr);
    await Block.insertIntoBlockTable(blockArr);

    //Parse Scenes
    let sceneJsonArr = video.insights_json.scenes;
    let sceneArr = Scene.parseSceneArr(sceneJsonArr);

    //Insert Scenes
    sceneArr = setBlockIdForAllScenes(sceneArr, blockArr);
    await Scene.insertIntoSceneTable(sceneArr);

    //Parse and insert Shots
    let shotJsonArr = video.insights_json.shots;
    let shotArr = Shot.parseShotArr(shotJsonArr);
    shotArr = setSceneIdForAllShots(shotArr, sceneArr);
    await Shot.insertIntoShotTable(shotArr);

    //Parse and insert KeyFrames
    let allKeyframesArr = [];
    shotArr.forEach(shot => {
      let keyframeJsonArr = shot.keyframe_json_arr;
      //Keyframe array for that particular shot
      let keyframeArr = Keyframe.parseKeyframeArr(keyframeJsonArr, shot, video);
      allKeyframesArr.push.apply(allKeyframesArr, keyframeArr);
    });
    await Keyframe.insertIntoKeyframeTable(allKeyframesArr);

    //Parse, Sort, Process and insert OCR data into the shot table
    let ocrJsonArr = video.insights_json.ocr;
    let ocrArr = Ocr.parseOcrJsonArr(ocrJsonArr);
    ocrArr.sort((ocr1, ocr2) => (ocr1.start_time > ocr2.start_time) ? 1 : (ocr1.start_time ==
ocr2.start_time) ? ((ocr1.top > ocr2.top) ? 1 : (ocr1.top == ocr2.top) ? ((ocr1.left > ocr2.left) ? 1 :
-1) : -1) : -1);
    ocrArr = setShotIdForAllOcrDetections(ocrArr, shotArr);
    ocrArr = cleanUpOcrDataWithinEachShot(ocrArr);
    let concatenatedOcrTextArr = concatOcrWithinSameShot(ocrArr);
    await Ocr.updateShotTableWithOcrData(concatenatedOcrTextArr);

    //Aggregate OCR data from the Shot table and insert it into the scene table
    for(let i=0; i<sceneArr.length; i++) {
      try {
        let currScene = sceneArr[i];
        let currSceneId = currScene.scene_id;
        let shotsOcrJsonArr = await db.any('SELECT original_ocr FROM shot WHERE scene_id = $1
ORDER BY shot_num ASC', [currSceneId]);
        let ocrArrForScene = [];
        for(let j=0; j<shotsOcrJsonArr.length; j++) {
          ocrArrForScene.push(shotsOcrJsonArr[j].original_ocr);
        }
        ocrArrForScene = Utils.removeDuplicatesFromStringArr(ocrArrForScene);
        let finalSceneOcr = '';
```

```

        for (let j=0; j<ocrArrForScene.length; j++) {
            if(ocrArrForScene[j] != '' || ocrArrForScene[j] != undefined || ocrArrForScene[j] !=
' ' || ocrArrForScene[j] != '\n') {
                finalSceneOcr = finalSceneOcr + '\n' + ocrArrForScene[j];
            }
        }
        finalSceneOcr = finalSceneOcr.trim();
        await db.one("UPDATE scene SET original_ocr = $1 WHERE scene_id = $2 RETURNING id",
[finalSceneOcr, currSceneId], a => + a.id);
    } catch(e) {
        console.log(e);
    }
}
await Scene.aggregateShotOcrIntoSceneOcr(video);

//Fetch captions for all keyframes and update the keyframes table
let imgAnalysisResult = await fetchImageAnalysisForAllKeyFrames(allKeyframesArr);
allKeyframesArr = imgAnalysisResult.allKeyframesArr;
await Keyframe.updateCaptionDataForKeyframes(allKeyframesArr);

//Assign a description to each shot by using caption data from keyframes
shotArr = assignDescriptionToEachShot(shotArr, allKeyframesArr);
await Shot.updateDescriptionDataForAllShots(shotArr);

//Assign a description to each scene by using description data from shots
sceneArr = assignDescriptionToEachScene(sceneArr, shotArr);
await Scene.updateDescriptionDataForAllScenes(sceneArr);

//Insert objects detected in each frame into the keyframe_object table
let keyframeObjectArr = imgAnalysisResult.keyframeObjectArr;
await KeyframeObject.insertIntoKeyframeObjectTable(keyframeObjectArr);

//Insert tags detected in each frame into the keyframe_tag table
let keyframeTagArr = imgAnalysisResult.keyframeTagArr;
await KeyframeTag.insertIntoKeyframeTagTable(keyframeTagArr);

//Fetch environment IDs for all keyframes and update the keyframes table
let kfEnvironmentArr = await fetchEnvironmentForAllKeyFrames(allKeyframesArr);
await Environment.insertIntoKeyframeEnvironmentTable(kfEnvironmentArr);
for(let i=0; i<blockArr.length; i++) {
    let currBlock = blockArr[i];
    let currBlockId = currBlock.block_id;
    // For each block, fetch the data inserted in the previous step
    let envDataForBlock = await db.any('SELECT distinct ke.environment_id,
(COUNT(ke.environment_id)*AVG(ke.probability)) as score, env.category_name FROM environment env,
keyframe_environment ke, keyframe k, shot sh, scene sc, block b, video v WHERE ke.environment_id =
env.environment_id AND ke.keyframe_id = k.keyframe_id AND k.shot_id=sh.shot_id AND sh.scene_id =
sc.scene_id AND sc.block_id = b.block_id AND b.block_id = $1 GROUP BY ke.environment_id,
env.category_name ORDER BY score DESC', [currBlockId]);
    // Set an environment description for the block
    currBlock.setOriginalDescription('The environment is either ' +
envDataForBlock[0].category_name + ' or ' + envDataForBlock[1].category_name);
    console.log(currBlock);
}
Block.updateBlockTableDescription(blockArr);
return true;
} catch(e) {
    console.log(e);
    return false;
}
}

```

5.3.2. Caption Generation

The PNG image files (each file representing a keyframe) produced in the first step are passed as input to the Azure Computer Vision API [21] which returns probable captions for the image along with a list of objects/labels (Even bounding-box data is available) detected. This generated caption is thus assumed to indicate a description of the shot to which this keyframe belongs. The captions for shots within a scene are aggregated together to form a short paragraph (This step also involves data cleaning wherein duplicate captions are removed)

The API can be accessed by sending a POST request to the ‘/vision/v2.0/analyze’ API endpoint. The image URL parameter is sent through the body of the POST request to the Microsoft server. The subscription key should also be included in the message header. Here, it is also important to note that the image URL needs to be kept public for this feature to work (Ensure that while uploading a video to the video indexer, it’s data is kept as ‘public’ so that it can be accessed over the internet). Moreover, the free tier version of the API throttles (rate-limits) us to making not more than 1 request per 3 seconds. If the 3 second rule is not followed, the API returns a 400 error (Bad Request).

The API returns metadata about the image (only the data types requested in the URL parameters are returned). The most important data that is collected is the generated caption for the image. This caption along with its corresponding confidence score are stored in the database (accompanied by other data such as objects detected, tags, etc.)

The image analysis API accepts parameters of 3 types:

- Visual Features: Such as Adult, Brands, Categories, Color, Description, Faces, ImageType, Objects, Tags
- Details: Such as Celebrities, Landmarks
- Language: Such as en (English), es (Spanish), fr (French), etc.

A sample URL:

```
https://westus.api.cognitive.microsoft.com/vision/v2.0/analyze?visualFeatures=Categories,Description&details=Landmarks&language=en
```

Code for image analysis (Contained in /backend/apirepository.js):

```
const imageAnalysisSubscriptionKey = '9a17df10db4542a6863bf2621e198916';
const MS_COGNITIVE_SERVICES_BASE_URL = 'https://westus.api.cognitive.microsoft.com/vision/v2.0/';

async function getImageAnalysis(imageUrl) {
  try {
    const imgAnalysisUrl = MS_COGNITIVE_SERVICES_BASE_URL +
      'analyze?visualFeatures=Adult,Brands,Categories,Color,Description,Faces,ImageType,Objects,Tags&details=Celebrities,Landmarks&language=en';
    const body = { url: imageUrl };
    return await fetch(imgAnalysisUrl, {
      method: 'post',
      body: JSON.stringify(body),
      headers: {
        'Content-Type': 'application/json',
        'Ocp-Apim-Subscription-Key': imageAnalysisSubscriptionKey
      },
    }).then(res => res.json());
  } catch(e) {
    throw e;
  }
}
```

5.3.3. Facial characteristics detection

This is a feature that was used in the initial phases of the project but remains discarded in its current state. However, it may be useful in future iterations of the project in use cases where the visually impaired user demands information about the on-screen character's facial features and expressions.

How it works: The JPG image files (each file representing a face) produced the previous step are passed as input to the Microsoft Azure Face API [22]. This service makes machine learning-based predictions of features and attributes, including age, emotion, gender, pose, smile, and facial hair for each face in the image along with the respective probabilities for each of these features.

The API can be accessed by sending a POST request to the '/vision/v1.0/detect' API endpoint. The image URL parameter is sent through the body of the POST request to the Microsoft server. The subscription key should also be included in the message header. Important parameters accepted by the facial analysis API include: age, facialHair, gender, smile, hair, emotion, accessories.

Facial analysis code sample from a previous version of the project:

```
const axios = require('axios');

const MS_COGNITIVE_SERVICES_BASE_URL = 'https://westus.api.cognitive.microsoft.com/vision/v1.0/';

async function getFaceAnalysisFromImage(faceThumbnailUrl) {
  let apiUrl = MS_COGNITIVE_SERVICES_BASE_URL +
    'detect?returnFaceId=false&returnFaceLandmarks=false&returnFaceAttributes=age,facialHair,gender,smile,ha
    ir,emotion,accessories&recognitionModel=recognition_02&returnRecognitionModel=false'
  try {
    let data = JSON.parse('{"url": "' + faceThumbnailUrl + '"}');
    const output = await axios({
      method: 'post',
      url: apiUrl,
      headers: {'Ocp-Apim-Subscription-Key': '620982a0d1d24a27bf98f384e390620f', 'Content-
      Type': 'application/json'},
      data
    });
    console.log(output.data);
    return output.data[0].faceAttributes;
  } catch(e) {
    console.log(e);
  }
}
```

5.3.4. Environment detection using the Places365 CNN model

The PNG image files (each file representing a keyframe) produced in the first step are passed as input to the Places365-CNN [23] model (IBM Max Scene Classifier) [13]. The API response contains information about whether it's an indoor or outdoor image. It is also capable of returning the environment category (The model is trained to detect 365 types of environments [23]). The environment detected for this keyframe is assumed to be the environment of the corresponding shot.

Source code for the CNN model: <https://github.com/IBM/MAX-Scene-Classfier>

To use this model, the following steps need to be followed:

- Build the model:
 - Install Docker (<https://www.docker.com/>)
 - Clone the IBM repository locally. In a terminal, run the following command:

```
git clone https://github.com/IBM/MAX-Scene-Classfier.git
```
 - Change directory into the repository base folder:

```
cd MAX-Scene-Classfier
```

- To build the docker image locally, run:
`docker build -t max-scene-classifier`
- Deploy the Model:
 - To run the docker image, which automatically starts the model serving API, run:
`docker run -it -p 5000:5000 max-scene-classifier`
- Use the Model:
 - Send a POST request to <http://0.0.0.0:5000/model/predict>. Include the image that needs to be analyzed as a readable stream within the form data.

The API returns the 5 environments with the highest confidence scores (environment IDs are returned - these IDs correspond to predefined 365 environments that the model has been trained on. The 'environment' table acts as a lookup table to get information about the ID such as environment category, whether it is indoor or outdoor, whether it's natural or manmade, etc.)

Code for Environment Detection (Contained in `/backend/apirepository.js`):

```
const IBM_MAX_SCENE_CLASSIFIER_URL = 'http://0.0.0.0:5000/model/predict';

async function getSceneClassification(imageUrl, frameId) {
  try {
    const imageSavePath = path.join(__dirname, '../' + frameId + '.png');
    await Utils.downloadFile(imageUrl, imageSavePath);
    const formData = new FormData();
    formData.append('image', fs.createReadStream(imageSavePath));
    return await fetch(IBM_MAX_SCENE_CLASSIFIER_URL, {
      method: 'POST',
      body: formData
    }).then(res => {
      fs.unlinkSync(imageSavePath);
      return res.json()
    });
  } catch (error) {
    throw error;
  }
}
```

5.3.5 Text to Speech Conversion

The Google Cloud Text-To-Speech API [14] is used for the conversion of OCR-text and caption-text into Audio descriptions (MP3 Files).

- All rows from the description table are pulled and for each row, the OCR text and description text are concatenated. This concatenated text is passed to the Google API which in turn returns an MP3 file.
- All the generated MP3 files are stored inside the '/public/audio/' directory within the file system. Within the 'audio' directory, the generated MP3 file is stored under a hierarchy of folder which looks like:
/video_id/ai_type/volunteer_id/scene_number.mp3.
- The default language for the output MP3 file is set as en-US (American English)

Code used for converting all descriptions in DB to Audio files (Contained in /backend/texttospeech.js):

```
async function generateMp3ForAllDescriptionsInDb() {
  try {
    let descriptionListArr = await db.any('SELECT * FROM descriptions order by id');
    for(let i=0; i<descriptionListArr.length; i++) {
      let currDescription = descriptionListArr[i];
      let currDescriptionId = currDescription.desc_id;
      let currVideoId = currDescription.video_id;
      let volunteerId = currDescription.volunteer_id;
      let hasAi = currDescription.has_ai;
      let descriptionText = currDescription.description;
      let ocrText = currDescription.ocr;
      let sceneNum = currDescription.scene_num;
      ocrText = (hasAi && ocrText && ocrText.trim().length>0) ? "\nText on screen is : " + ocrText
: "";

      let finalText = descriptionText + ocrText;
      let descType = hasAi ? 'ai' : 'non_ai';
      let fileNamePath = `public/audio/${currVideoId}/${descType}/${volunteerId}/${sceneNum}.mp3`;
      let relativePath = fileNamePath.replace('public', '..');
      await generateMp3ForTextInFilesystem(fileNamePath, finalText);
      let duration = await getAudioDurationInSeconds(fileNamePath);
      console.log(duration);
      await updateDescriptionTable(duration, currVideoId, volunteerId, hasAi, sceneNum,
relativePath);
    }
    return true;
  } catch (error) {
    console.log(error);
    return false;
  }
}
```


5.4 User Study Phase I - Describer (Volunteer) Module and User Interface

This is the final UI that is shown to volunteering describers. The descriptions (with the segmented timings) generated in the previous step are displayed as editable fields in a scrolling list where each description textbox within the list is synchronized to certain timestamps in the video. The volunteer may choose to modify the AI generated descriptions and modify the OCR text in a way that he/she feels would be more beneficial for a blind or visually impaired user of YouDescribe. The AI generated descriptions and OCR text are fetched from the PostgreSQL database via a REST API written using Node.js + Express.js backend server.

5.4.1 Home Page

- On clicking on any of the participants on the list, the page redirects to video list page.
- Developed the logic for routing the user to their respective video list page.
- Backend APIs called:
 - `/userStudy/users/` (GET) - Fetches list of all participants from the `us_user` table

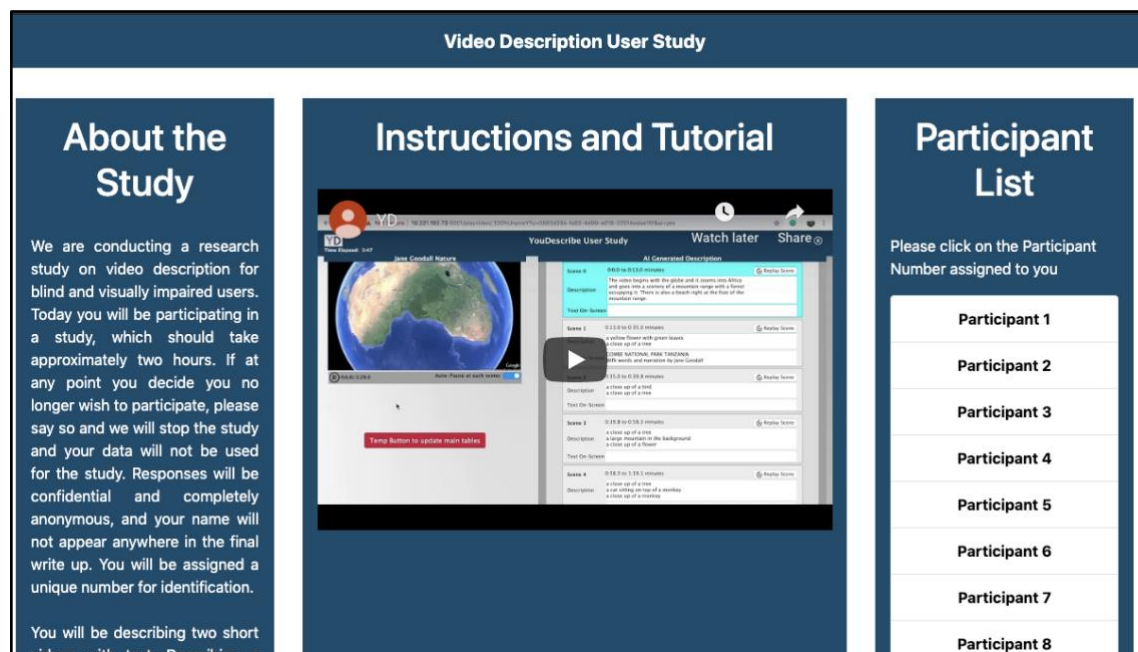


Figure 3: User Study Phase I 'Homepage' Screenshot

5.4.2 Video List page

- Created a page which consist of two videos selected for each user.
- The videos displayed are in a predefined order, one with AI support and other video without AI support.
- Each video also has links corresponding to a survey(Google Form) that the user is supposed to fill out after submitting a video description.
- Backend APIs called:
 - `/userStudy/users?userId=[user_id]` (GET) - Fetches list of Videos for that particular participant in JSON format. Each JSON object contains metadata such as video_id, title and a boolean to indicate whether AI support is enabled for that video.

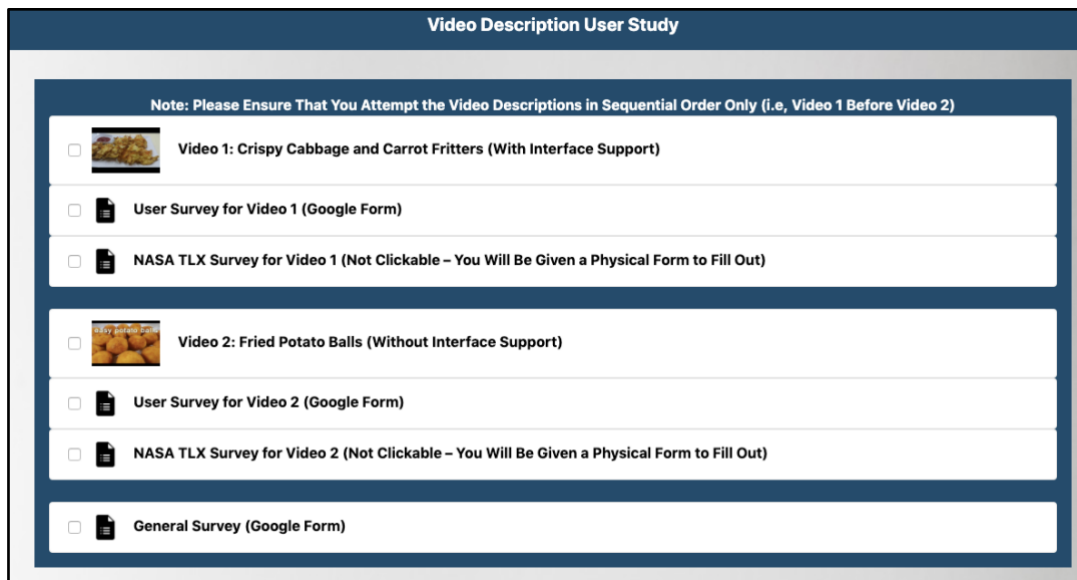


Figure 4: User Study Phase I ‘Video List Page’ Screenshot

5.4.3 Video Description Page without AI assistance

This page consists of a scrollable UI where the end-user (volunteer describer) is supposed to manually decide the number of scenes as well as their start times. An empty input box is provided to input the description and OCR detected within that scene.

- Backend APIs Called:
 - `/fetchVideoData?videoid=[youtube_id]` (GET) - Fetches all metadata for the video under consideration from the backend. However, in this case only the 'video_title' metadata is used.
 - `/saveNonAiDescription` (POST) - Submits the scenes and descriptions created by the volunteer to the backend which in turn saves this information into the PostgreSQL database.

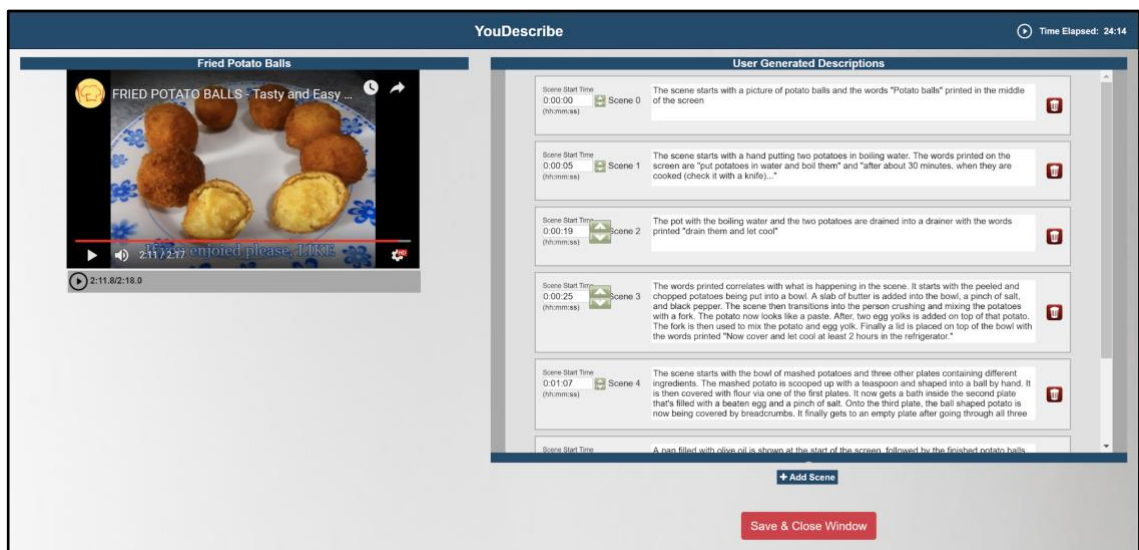


Figure 5: User Study Phase I ‘Video Description Page without AI assistance’ Screenshot

5.4.4 Video Description Page with AI assistance

- This page consists of a scrollable UI where the end-user (volunteer describer) is provided with a list of scene segments that have been determined by the AI. Each scene segment consists of textboxes for on screen text as well as for scene description. The machine generated captions and detected OCRs are fetched from the PostgreSQL database using API calls to the NodeJS backend.

- JavaScript functions were written to trigger scrolling of the description and OCR text boxes at specific timestamps corresponding to scene start-times. Multiple JQuery and Bootstrap.js functions were used to give the desired effect. Auto-pausing of the video at the beginning of each scene was also implemented - it can be toggled on/off. The ability to replay a screen was also a feature that was added in order to improve usability.
- Backend APIs called:
 - `/fetchVideoData?videoid=[youtubeId]&userId=[userId]` (GET) - Fetches video data from the backend such as scene and block level descriptions. In its current form, data is fetched from the 'scene' table. However, in future versions it should be fetched from the 'scene_modification' table. The flaw with the current implementation is that when the page is reopened, it does not reflect the changes made by the describer because the modified descriptions are not being fetched even though they are saved in the database.
 - `/saveAiDescription` (POST) - Submits the descriptions and OCRs updated by the volunteer to the backend which in turn saves this information into the PostgreSQL database.

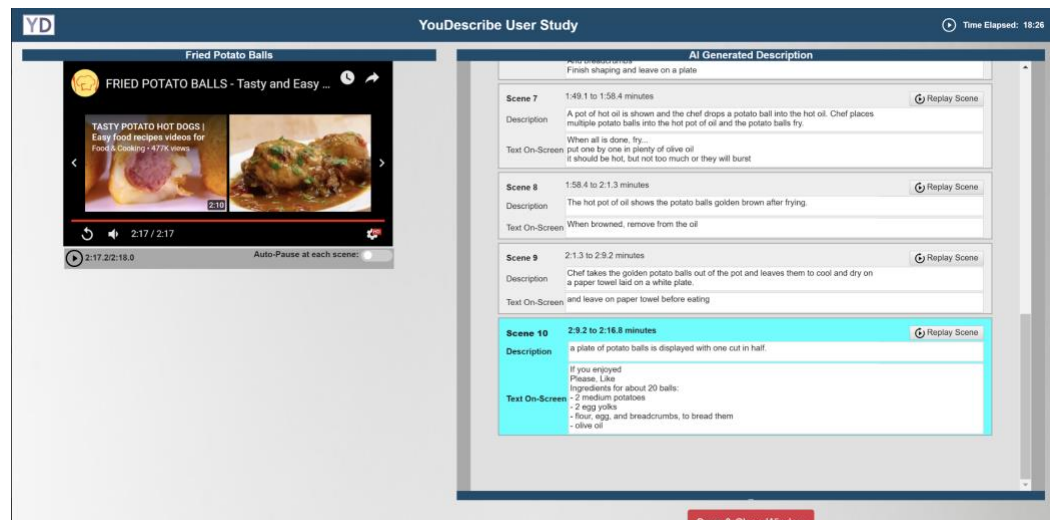


Figure 6: User Study Phase I ‘Video Description Page with AI assistance’
Screenshot

5.5 User Study Phase II

5.5.1 Home Page

- The home page consists of a list of participants.
- Backend APIs called:
 - `/userStudy2/users (GET)` - Fetches list of users for the second user study from the database.

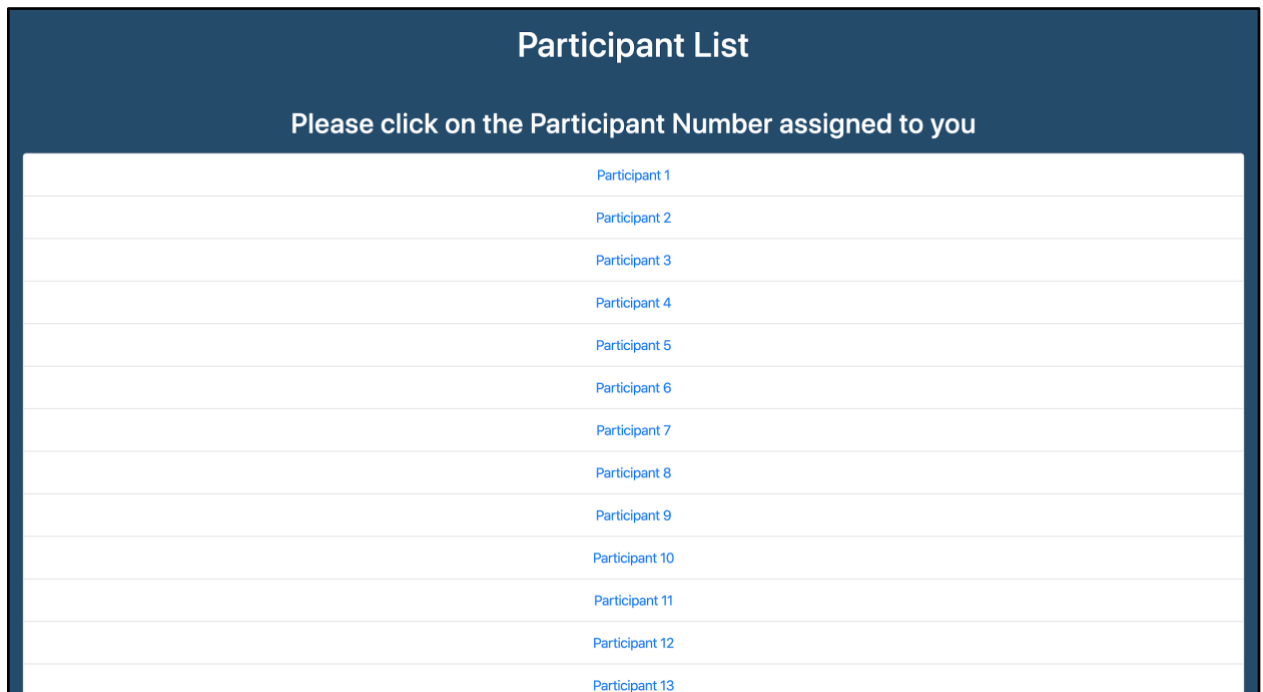


Figure 7: User Study Phase II ‘Home Page with Participant List’ Screenshot

5.5.2 Video List & Rating Page

- This page consists of 22 video links which lead to another page containing the corresponding videos along with their respective audio descriptions.
- Participants have the ability to rate the quality of descriptions contained within each video link from this page.
- The page was developed for blind and visually impaired users who read the screen using a screen reader. Therefore, special attention was paid in terms of accessibility. Every option, button provided on the page had ARIA labels.

- The user interface consists of a list of video-links. Each link is accompanied by a section for rating the video out of 5 (1 being poor and 5 being excellent) and a text box for recording any comments/observations the user would like to share.
- The page had two general survey questions regarding video descriptions, which was further used in the user study statistics.
- Backend APIs called:
 - **/userStudy/helper/video?userId=[user_id] (GET)** - Fetches a list of videos that have to be shown for this particular participant (the participant is identified by the user ID parameter)
 - **userStudy/getUserHelper?userId=[user_id] (GET)** - Retrieves any preferences and information saved previously by user such as:
 - Image access preference: description reliance, magnification, tactile graphic feedback
 - Video description frequency
 - Email Address
 - **/updatefeedback (POST)** - Submit rating and comment values to the backend for each individual video (accompanied by other metadata such as video ID, volunteer ID, AI support enablement). This data is posted to the backend on change of focus (i.e., on pressing the tab button or clicking the mouse elsewhere).
 - **/userStudy/helperStatus?userId=[user_id] (GET)** - Used to check whether a rating has been given to all the assigned videos by the visually impaired user.
 - **/userStudy/updateStatus (PUT)** - Updates the status of the 'video description rating' task at the backend for this particular user. The request body contains the user ID as well as the status. Once the user has rated all the videos assigned to him/her and presses the 'Done' button, the status is marked as completed in the database.
 - **/userStudy/updateUserInfo (PUT)** - Used to update user email, image access preferences and video description frequency. Triggered whenever the corresponding fields(checkboxes) are clicked.

Video Description User Study

Thank you for participating in this user study. Please answer the questions below, and then watch the videos, and answer the questions about each video. There are 22 videos, and each video is about 2 minutes long. The 22 videos are in fact only 2 different videos described by 11 different describers. Even though they are repetitive, please listen to each description, rate the description quality, and leave comments for each video. At the end, there is a button confirming that you completed all the videos. For each video, there is a link to the video on YouTube. Open the link and click the play button. The YouTube video will be played along with the audio description.

Please play each video once all the way through to the end (there are no pause or rewind functions). You are welcome to take a break in between videos. If the Youtube video fails to load, please click the refresh button on your browser. Thank you again for participating.

Please enter your email address:

How do you typically prefer to access images?

☒ With magnification

☐ Rely on description

☐ Tactile graphic

☐ Other :

How regularly do you use video descriptions?

☒ Never

☐ Very infrequently (a few times a year)

☐ Sometimes (a few times a month)

☐ Very Frequently (several times a week)

☐ Almost daily or daily

Figure 8: User Study Phase II ‘Video List & Rating Page’ Screenshot 1

Comment:



Video 22: Fried Potato Balls

Please rate the description of this video on a scale of 1 to 5, with 1 being the poorest and 5 being the best:

☐ 5 - Excellent

☐ 4 - Good

☐ 3 - Average

☐ 2 - Fair

☐ 1 - Poor

I feel I have a good understanding of the topic covered in this video due to the video.:

☐ Strongly Agree

☐ Agree

☐ Neutral

☐ Disagree

☐ Strongly Disagree

Comment:

Done and send me Amazon Gift Card!

Figure 9: User Study Phase II ‘Video List & Rating Page’ Screenshot 2

5.5.3 Video + Audio Description Page

- This page has an HTML Iframe consisting of the youtube video that has been setup for playback.
- In addition, an audio description of the video that is synchronized with the video can be heard by the blind/visually impaired user. This audio description corresponds to the textual description given by volunteers(participants) in Phase 1 of the user study.
- Backend APIs used:
 - `/fetchVideoData?videoid=[video_id]` (GET) - Fetches information about the video such as title, duration, scenes, start/end times of scenes.
 - `/audio/[video_id]/[type_ai_non_ai]/[volunteer_id]/[scene_num].mp3` (GET) - Fetches MP3 audio files from the backend that correspond to that particular scene by a specific volunteer.



Figure 10: User Study Phase II ‘Video + Audio Description Page’ Screenshot

5.6 YouTube IFrame API

The YouTube IFrame player API lets has been used to embed a YouTube video player on the webpage and control the player using JavaScript functions. After the API's JavaScript code loads, the API calls the 'onYouTubeIframeAPIReady' function, at which point a YT.Player object was constructed to insert a video player on the webpage.

YouTube IFrame API configuration:

```
function onYouTubeIframeAPIReady() {
  player = new YT.Player('player', {
    height: '270',
    width: '480',
    videoId: 'ZH7V2tU3iFc', // Youtube ID of the video to be played
    playerVars: {
      autoplay: 1,          // toggle automatic video playback when player loads
      enablejsapi: 1,       // enables the player to be controlled via JS API
      cc_load_policy: 1,    // Enable or disable auto-display of closed captions
      controls: 1,          // Toggle Video Player Controls Visibility
      fs: 0,                // toggle display of 'full screen' button
      iv_load_policy: 3,    // video annotations not shown by default if value=3
      modestbranding: 1,   // prevents the display of Youtube logo if value=1
    },
    events: {
      'onReady': onPlayerReady, // Called once the video is loaded
      'onStateChange': onPlayerStateChange // When state changes from between pause, play,
      buffering, ended, started, ended, etc.
    }
  });
}
```

6. Evaluation

6.1. User Study - Phase 1: Describing Videos

In Phase 1 of the user study, the focus was on determining the effectiveness of the HILML system in the process of description generation for novice describers. Participants of the study were made to describe videos in two conditions: one with the HILML system with machine learning assistance for text generation and scene segmentation, and one with no machine learning assistance.

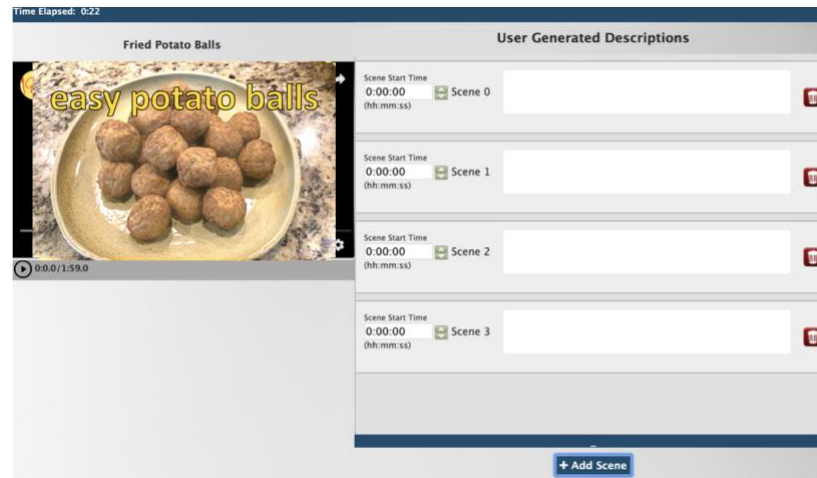


Figure 11: User Interface for the sighted volunteers in the Human-Only Panel

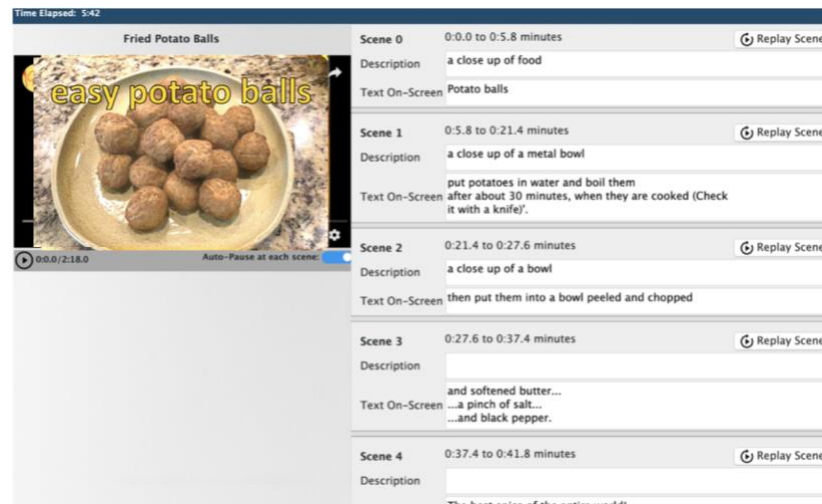


Figure 12: User Interface for the sighted volunteers in HILML conditions

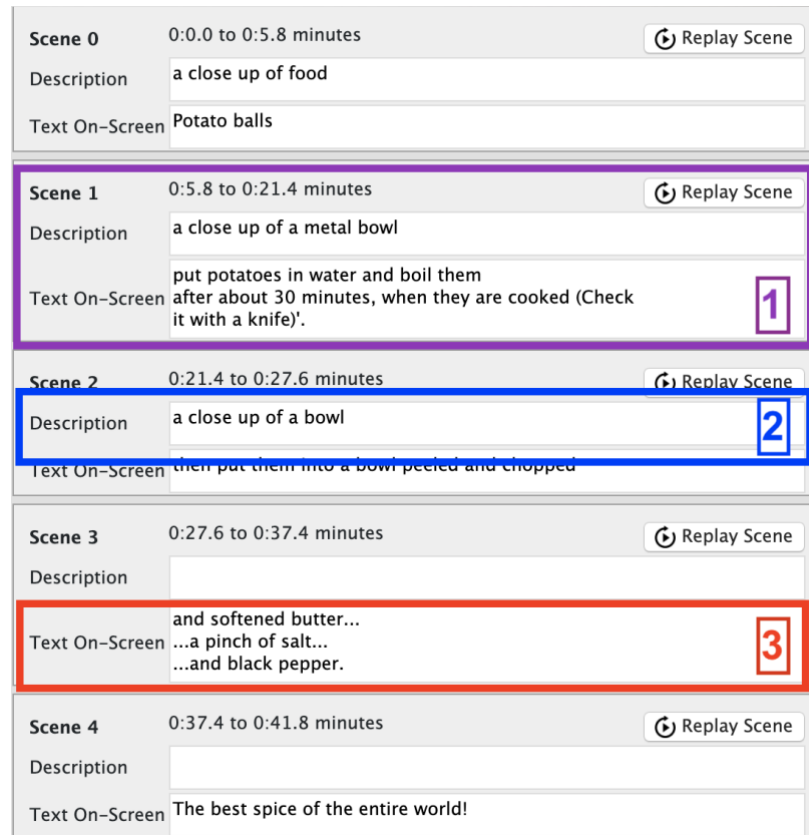


Figure 13: Close-up of the Describer User Interface and script editor of the HILML system displaying: 1) automated scene segmentation, 2) text generation for the description of images in the video, and 3) text generation for text that appears on the screen. [27]

6.1.1 Experimental Design for Phase 1 of the User Study

Twenty-two participants were called in for the user study. All were asked to go through a short tutorial to acquaint themselves with audio description fundamentals and the software. Each participant was then given two videos to describe, one with and one without the HILML system.

The two types of systems can be described as:

- **HILML (AI Assisted) System:** Automated textual descriptions and scene segmentation of the target video were provided to the describer. The describers could then choose to add, modify or remove descriptions if they wished to do so.

- **Non-HILML (Manual) System:** In the ‘Human-Only’ condition, participants typed out all of their video descriptions from scratch in the user interface.

All the participants were given two ‘How-To’ cooking videos of around 2 minutes in length.

Both videos were ‘How-To’ cooking videos of around 2 minutes length each. This is one of the most common video types requested by users on the Youdescribe website.

The describers were divided into two groups: Group 1 described Video 1 with the HILML system and Video 2 without the HILML system. Group 2 described Video 1 without the HILML system and Video 2 with the HILML system. After finishing each description task, the participants were asked to fill out a questionnaire to assess their experience of using the provided system. Participants were also asked to fill out a NASA-TLX survey - a workload assessment tool [24].

6.1.2 Results for Phase 1 of the User Study

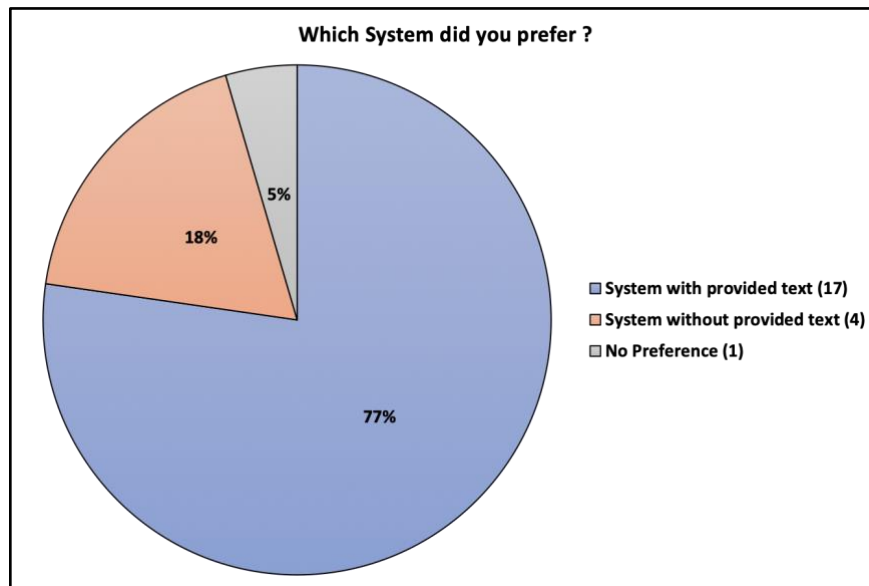


Figure 14: Majority of describers preferred the HILML System with provided text

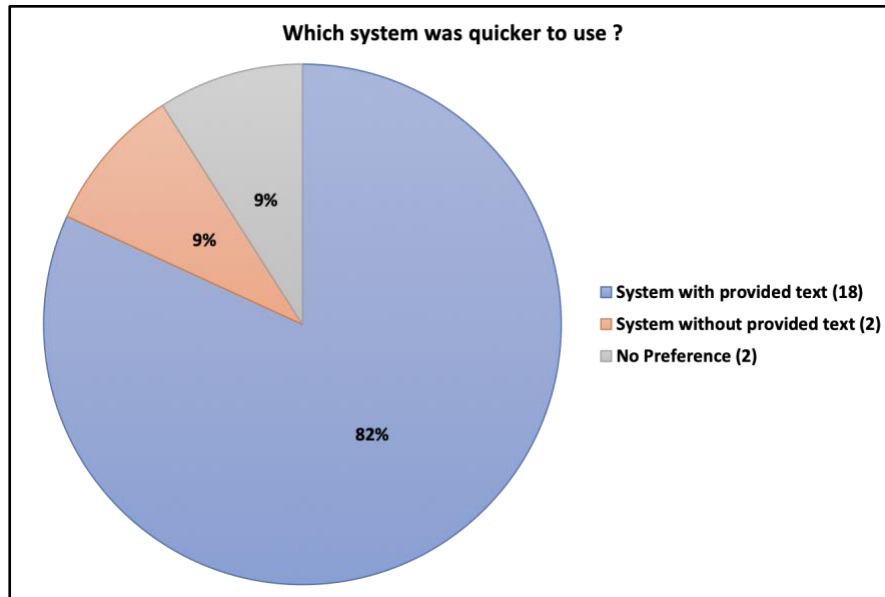


Figure 15: According to a majority of describers, the HILML System with provided text seemed quicker to use

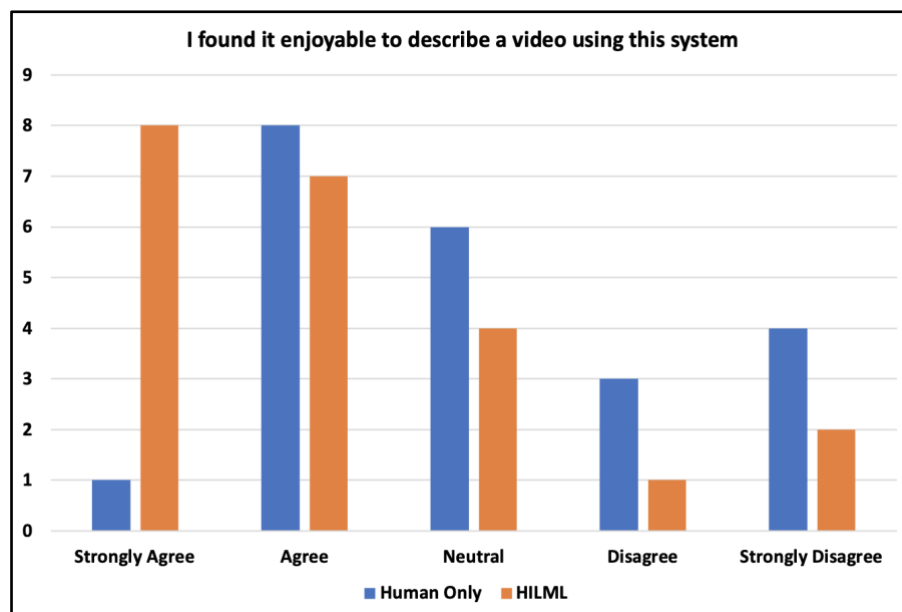


Figure 16: HILML vs Human Only condition. Parameter: 'Enjoyability' [27]

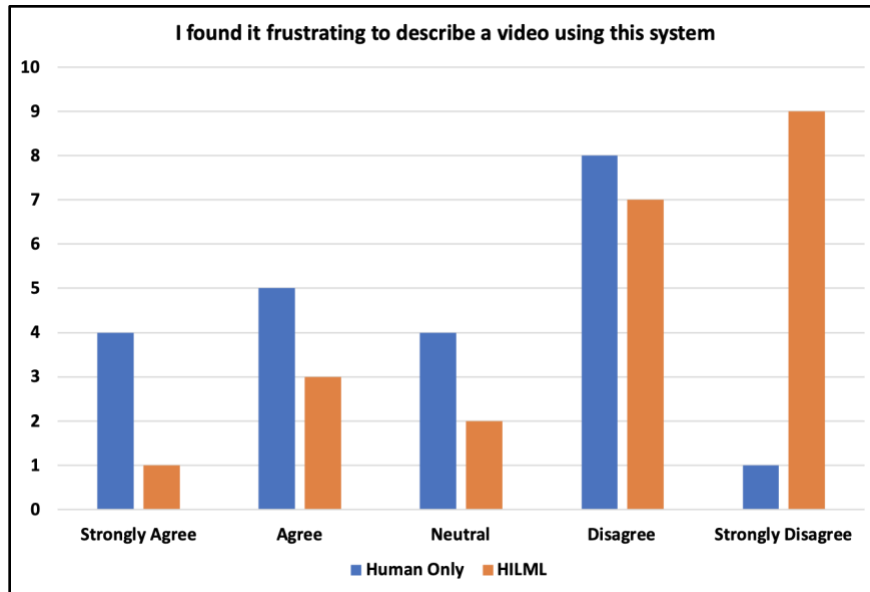


Figure 17: HILML vs Human Only condition. Parameter: ‘Frustration’ [27]

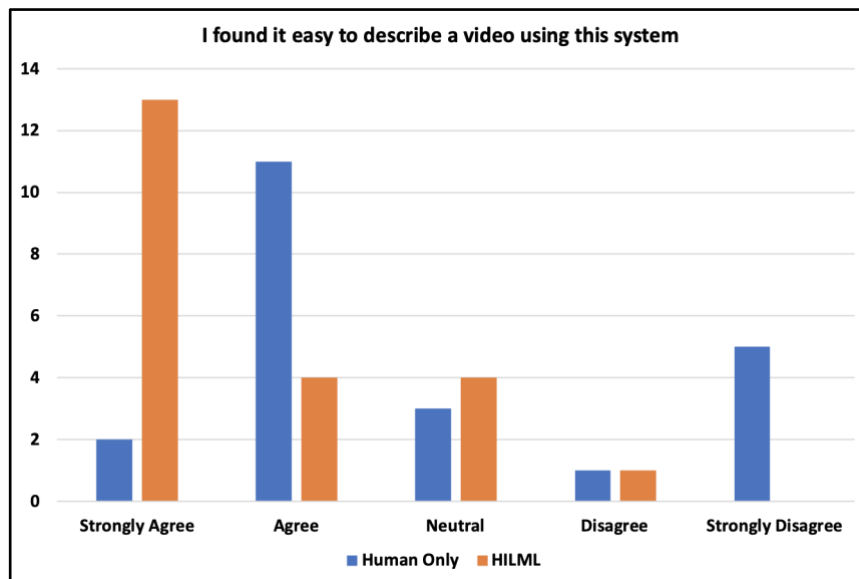


Figure 18: HILML vs Human Only condition. Parameter: ‘Ease of Use’ [27]

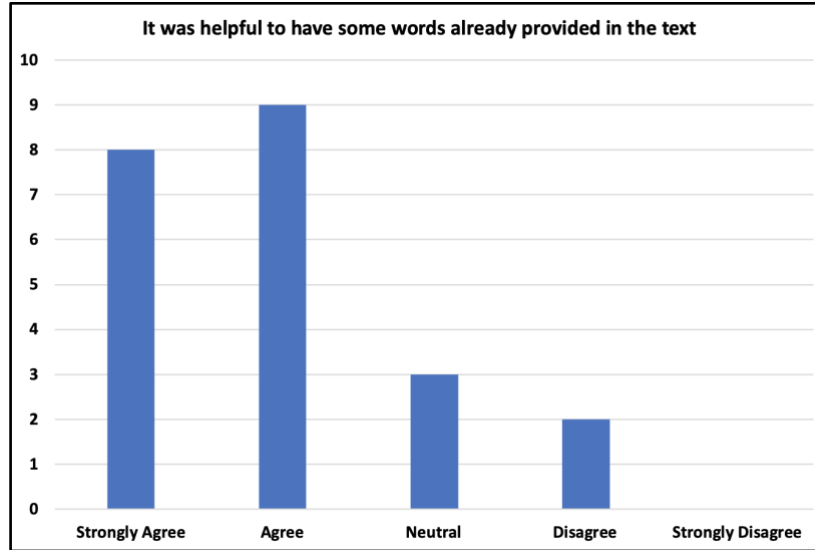


Figure 19: Describers' ratings of the helpfulness of the provided text generated by the HILML system [27]

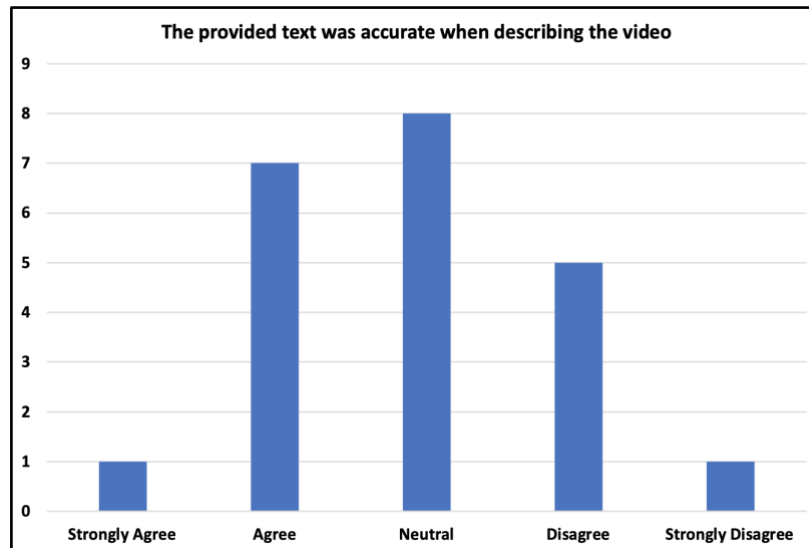


Figure 20: Describers' ratings of the accuracy of the provided text generated by the HILML system. The most highly rated level was 'neutral' [27]

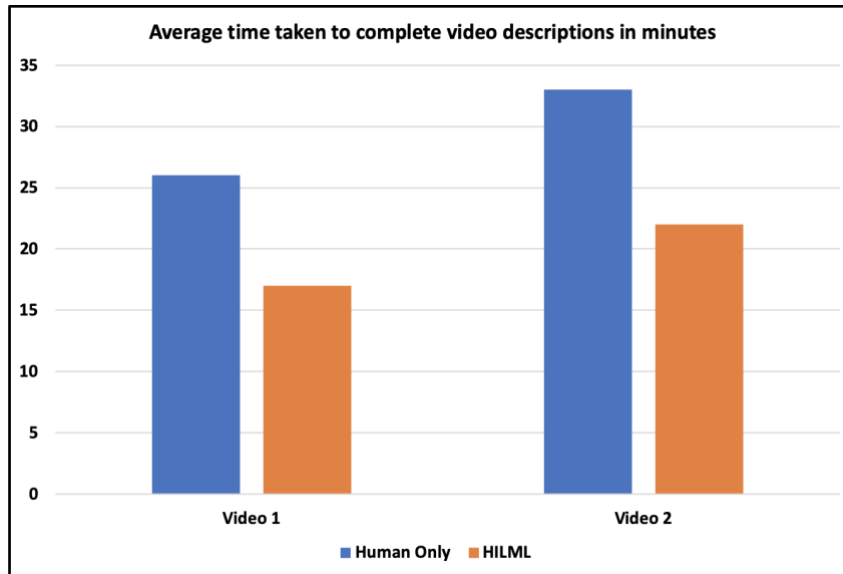


Figure 21: Average time taken to complete the video descriptions between the HILML and Human-Only conditions.

6.1.2.1 Questionnaire Data

Questionnaire data shows that describers preferred the HILML system and felt that it was quicker than the Human-Only condition even though both systems were generally easy to use for most participants. This is interesting as the accuracy of the provided text by the HILML system was reported as being somewhat accurate, with the mode response being 'neutral'. This suggests that even though the accuracy of the HILML system was not perfect, it was still helpful than no machine learning assistance at all.

6.1.2.2 Time Taken to Complete Video Descriptions

Results showed that users described videos with the HILML condition significantly faster than the Human-Only condition. These findings correspond with the questionnaire data by describers that the HILML system was quicker to use than the Human-Only condition.

6.1.2.3 Describer Workload

	Human-Only	HILML	Wilcoxon Z	<i>p</i>	effect size
Mental demand	64.1 (20.7)	43.4 (23.9)	-2.486572	0.011110	0.530139
Physical demand	19.8 (20.5)	14.8 (17.6)	-0.813411	0.427694	0.173420
Temporal demand	40.5 (29.7)	28.0 (22.0)	-1.463076	0.148385	0.311929
Performance	31.1 (22.3)	26.8 (27.8)	-0.179176	0.866728	0.038200
Effort	60.7 (18.5)	45.7 (23.3)	-2.199999	0.026089	0.469041
Frustration	43.4 (27.3)	23.6 (22.9)	-2.548356	0.008972	0.543311

Table 3: Results of the NASA-TLX responses - Mean (St. dev.) NASA-TLX values (0=low, 100=high) and results from Wilcoxon Signed-rank test. Rows in bold indicate significant results. [27]

Results of the NASA-TLX responses (Table 3) [27] showed that describers found the HILML system to be significantly less mentally demanding, require less effort, and be less frustrating. This is consistent with the subjective questionnaire data that describers found the HILML system to be easier to use, less frustrating, and more enjoyable than the Human-Only condition.

6.2. User Study - Phase 2: Listening and judging audio description quality

In Phase 2 of the user study, the focus was on determining the quality of descriptions generated in the previous phase. Blind and visually impaired users were asked to participate in the study and give feedback regarding the quality of audio descriptions synchronized with 2 specific videos.

6.2.1. Experimental Design for Phase 2 of the User Study

All raters worked remotely and were each given half of the 44 videos described (as each of the 22 describers had produced 2 videos). Six raters were given videos produced by the first 11 describers and six raters were given videos described by the last 11 describers. Therefore, half of the videos provided to each rater were described using the HILML system and half were described by the Human-Only condition. The videos were split evenly into Video 1 and Video 2. The order of the videos provided to raters were randomized. Raters did not know which videos had been produced by the HILML system or by the Human-Only condition. They were not even aware that some videos had been described using the assistance of machine learning technology, they were simply told that the videos had been described by 11 different describers. They were not told which pairs of videos were described by the same describer.

Raters could play each video through and hear the video descriptions being read out by text-to-speech synthesis. After watching each video, raters were asked to rate the quality of description on a scale of 1 to 5 ('Poor' to 'Excellent'), their understanding of the topic in the video, and any other comments they had about the video.

6.2.2 Results for Phase 2 of the User Study

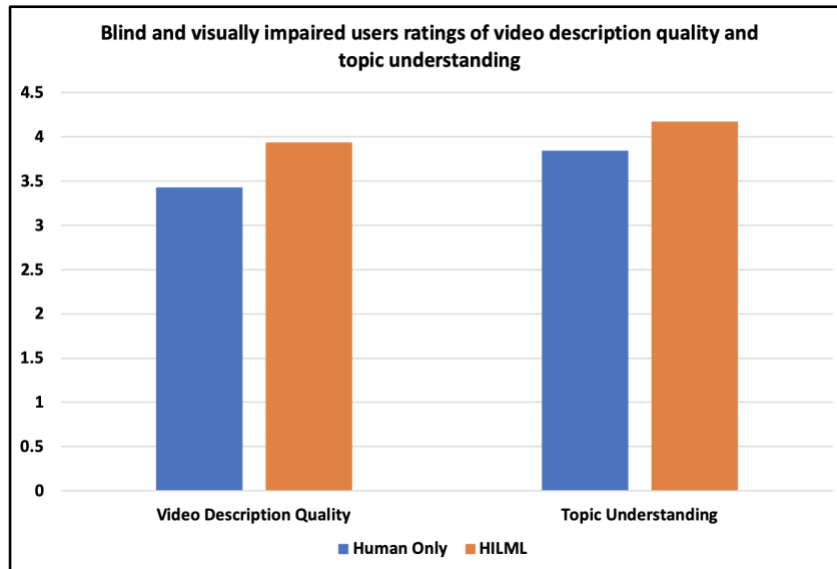


Figure 22: Average ratings [Likert Scale 1 (Poor) -5 (Excellent)] made by blind and visually impaired users on the quality of video descriptions and understanding of topic in videos described by the HILML and Human-Only conditions

6.2.2.1 Video Description Quality and Topic Understanding

Wilcoxon Signed-ranks tests showed that mean video description quality was rated significantly higher in videos created by the HILML system than the Human-Only condition and that mean understanding of video's topic was rated significantly higher in videos created by the HILML system than the Human-Only condition.

6.3. Discussion

6.3.1 Positive outcomes of the user study

- The quality of the video description was rated significantly higher when created by the HILML system compared to the Human-Only control condition, and this is backed-up by comments from raters
- Scene segmentation created by the HILML system was highly beneficial to the describers as can be noted from survey results with most volunteers giving

positive feedback with regards to the predefined scene segments. Moreover, descriptions created by the HILML system were also rated significantly higher by the blind users - this is a testament to the effectiveness of the system as it was appreciated by both sets of users (volunteer describers and blind/visually impaired users).

- Describers reported significantly lowered cognitive workload while using the HILML system in addition to being able complete the description task faster. This statement can be backed up by statistical analysis of the surveys given to the describers and also the NASA-TLX subjective workload assessments.

6.3.2 Limitations

- For the sake of this user study, 2 specific videos were chosen - both these videos had music in the background, significantly high occurrence of on-screen text and no speech or dialogue. The above-mentioned conditions made the videos a bit easier to describe for the volunteering describers. This is one serious limitation of the user study as it does not test the efficiency of the system when the video type is relatively more complex.
- The accuracy of OCR (Optical character recognition) detection was rated as mostly neutral by the participants. The technology for this detection is far from perfect and is still a work in progress. This is further complicated by the plethora of post-production effects in professionally made videos which end up confusing the text-detection algorithms and APIs. Effects such as fading in and out of text are a primary cause for duplicate and faulty detections.
- In its current form, the caption-generation is not as good as it needs to be. The descriptions tend to be very short and vague. This also a serious limitation that needs to be overcome. The 'Future Work' section of this document lists possible ways in which this can be mitigated.

7. Conclusion

A Human-in-the-Loop Machine Learning (HILML) approach to automatically generate video descriptions and scene segmentation for online videos was successfully designed using a wide variety of public APIs and machine learning models. The corresponding website and backend were developed and deployed on a remote server. Human describers were able to efficiently edit the automated text generated by machine learning algorithms. The newly developed HILML system ensured increased video accessibility for blind users by making it easier and faster for novice describers to generate high quality descriptions.

In the first phase of experimentation, the HILML system was compared to a Human-Only control condition with no AI assistance for novice describers. Results showed that a vast majority of volunteers (describers) preferred the HILML system as it was significantly faster and easier to use. During in-person interviews, describers pointed towards features like automated scene-segmentation and on-screen textual detection as the primary reasons for preferring the HILML system.

In the second phase of experimentation, the blind/visually-impaired users rated the descriptions generated in the previous phase by volunteers (describers). Results showed that they rated the audio descriptions generated by the HILML system to be of significantly higher quality.

8. Future Work

The project team is currently working on improving the accuracy and quality of the AI generated descriptions. To achieve this, the discrepancies between the original descriptions (machine generated) and the revised descriptions (after modification by human volunteers) are being recorded and the revised versions of the descriptions are being used to retrain the caption generation model. In addition to the above, possibilities are being explored to train a visual storytelling model to output a more appropriate and thorough description of individual frames and scenes within the video.

Another direction in which work has started is to develop a system that includes ‘on-demand descriptions’ for the blind and visually impaired users. The system is envisioned to allow a user to pause a video to ask questions using natural language (regular speech) and in response, the system should be able to answer the query. To enable the ‘on-demand descriptions’ feature, the plan is to train ‘visual question answering’ [25] and ‘visual dialog’ [26] models so that these models are able to respond and hold a meaningful dialog with blind users about the content of the video.

The team also plans to integrate the above-mentioned systems into the main YouDescribe website. Once the integration is done, we will investigate the extent to which we are able to reduce the barrier-to-entry for volunteer describers on the Youdescribe website (Our results from Phase 1 of the user study show that the HILML system makes it easier for untrained volunteers to come up with higher quality descriptions with lesser effort - which in turn should encourage greater participation). This should eventually lead to greater footfalls on the Youdescribe website and should have real-world impact as far as improving video accessibility to blind users is concerned.

Eventually, we should be able to generate consistent, well-structured data that will aid in the advancement of the field of video understanding as newer algorithms may be developed on the basis of this data. This dataset should be released under a free, open-source license so that it may be freely used by all machine learning researchers interested in solving similar problems.

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Increasing Video Accessibility for Visually Impaired Users with Human-in-the-Loop Machine Learning

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Abstract

Video accessibility is crucial for blind and visually impaired individuals for education, employment, and entertainment purposes. However, professional video descriptions are costly and time-consuming. Volunteer-created video descriptions could be a promising alternative, however, they can vary in quality and can be intimidating for novice describers. We developed a Human-in-the-Loop Machine Learning (HILML) approach to video description by automating video text generation and scene segmentation while allowing humans to edit the output. Our HILML system was significantly faster and easier to use for first-time video describers compared to a human-only control condition with no machine learning assistance. The quality of the video descriptions and understanding of the topic created by the HILML system compared to the human-only condition were rated as being significantly higher by blind and visually impaired users.

Author Keywords

Video Accessibility; Video Description; Blind Users; Visually Impaired Users; Machine Learning; Human-in-the-Loop;

CCS Concepts

•Human-centered computing → Accessibility technologies; Accessibility systems and tools;

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In this paper, we use the terms blind or visually impaired (BVI) users to refer to individuals who rely on audio descriptions, tactile graphics, or use magnification to typically access images.

Introduction

The World Health Organization (WHO) estimates that roughly 285 million people worldwide are visually impaired, and 39 million people are blind [3]. Despite international guidelines and standards [19, 6], there is still a paucity of videos on the Internet that are made accessible through video description for blind or visually impaired (BVI) users. Video description can be defined as “narration added to the soundtrack to describe important visual details that cannot be understood from the main soundtrack alone” [2]. A significant bottleneck in video accessibility for BVI users is the time and cost to produce video descriptions professionally. In this paper, we turn to the use of Human-in-the-Loop Machine Learning (HILML) to facilitate easier video description for novice describers and to improve the quantity and quality of video accessibility for BVI users. We built a video description interface with automated scene segmentation, text generation of images, and text generation of text on screen using machine learning. The human is then able to edit the generated output, allowing human-machine collaboration to produce high quality video descriptions while keeping a low barrier to entry for volunteer describers. We evaluated this interface on novice describers and compared it to a control condition with no machine learning assistance (Experiment 1). We then asked BVI users to rate the quality and clarity of the video descriptions produced by the two conditions (Experiment 2).

Related Work

Video Description Tools

LiveDescribe was designed for real-time descriptions of live television broadcasts with two automated features to work with silent periods. Feedback from describers was positive but also reflected the high cognitive workload of live description [4]. Villamizar et al. developed a prototype for an adaptive video enrichment system personal-

ized to BVI users based on decision trees [22], however, as stated by the authors, the prototype needs to be evaluated with user studies to investigate the adaptive system further [22]. Gagnon et al. [9, 8] built a prototype software for professional describers that used computer vision to extract visual content from videos. They displayed detected features on an interactive timeline which allowed professional describers to have an overview of the frequency and duration of features in the video, as well as the silent areas. Kobayashi et al. created audio description synthesis by using external metadata of videos on an online platform [14, 15]. Kobayashi et al. [15] developed a describer script editor with a visual interface to edit the video description sentences. They tested it on one novice and one professional describer. Ratings by blind and visually impaired users showed significantly lower scores for the novice describer [15]. The novice describer estimated that it would take three times longer without the script editor to produce video descriptions, but no control condition was carried out.

Machine Learning in Video Understanding

Classical video description approaches combined subject, object, and verb (SVO) detection from visual entities with template based language models to generate sentences [16]. However, the release of large datasets revealed that these methods cannot cope with the diversity in unconstrained open domain videos and were replaced with deep learning. In particular, Convolutional Neural Networks (CNNs) [1] are the state of the art for modeling visual data recognition [1, 24, 26] and Long Short-Term Memory (LSTMs) [12] are now dominating the area of sequence modeling such as NLP [5, 10, 25]. CNN, Recurrent Neural Network (RNN), or LSTM are used in the encoding stage to learn visual features that are then used in the second stage for text generation (decoding stage). For decoding, different flavours of RNNs are used, such as deep RNN,

Scene 0	0:0.0 to 0:5.8 minutes	⏮️ Replay Scene
Description	a close up of food	
Text On-Screen	Potato balls	
Scene 1	0:5.8 to 0:21.4 minutes	⏮️ Replay Scene
Description	a close up of a metal bowl	
Text On-Screen	put potatoes in water and boil them after about 30 minutes, when they are cooked (check it with a knife).	1
Scene 2	0:21.4 to 0:27.6 minutes	⏮️ Replay Scene
Description	a close up of a bowl	2
Text On-Screen	then put them into a bowl, pepper and chopped	
Scene 3	0:27.6 to 0:37.4 minutes	⏮️ Replay Scene
Description		
Text On-Screen	and softened butter... ...a pinch of salt... ...and black pepper.	3
Scene 4	0:37.4 to 0:41.8 minutes	⏮️ Replay Scene
Description		
Text On-Screen	The best spice of the entire world!	

Figure 1: Close-up of the Descriptor User Interface and script editor of the HILML system displaying: 1) automated scene segmentation, 2) text generation for the description of images in the video, and 3) text generation for text that appears on the screen. This automated text can then be edited by the describer (the text shown here is the automated output with no human edits).

Bi-directional RNN, LSTM, Gated Recurrent Units (GRU), Attention or Transformer [27]. The resulting description can be a single sentence or multiple sentences. Roem-mele et. al used a RNN architecture to generate stories in a sequence-to-sequence manner [21]. Martin et. al proposed an event representation for neural-network-based story generation [17]. Fan et al. created a hierarchical model that automatically generates stories conditioning on the writing prompts [7]. Huang et. al introduced the visual storytelling task, in which the trained model takes a sequence of photos as input and generates a short story that narrates this photo sequence [13].

Human-in-the-Loop Machine Learning

Human-in-the-Loop Machine Learning (HILML) has been defined as human and machine learning (ML) processes interacting to solve one or more of the following: 1) Making ML more accurate, 2) Getting ML to the desired accuracy faster, 3) Making humans more accurate, and 4) Making humans more efficient [18]. HILML is an important field to both the HCI and ML communities and can create important collaborations. In the field of image and video classification, Wang et al. [28] used HILML to create a semi-automatic method to segment foreground moving objects in surveillance videos. They combined human outlining of a small number of moving objects with a CNN to reach similar levels of accuracy as a human with less manual work than human-only analysis [28]. Pirrung et al. [20] created a HILML image organization web application where humans can organize their images and the machine learns from these groupings. In return, the machine can reposition images or regroups them to reflect its assessment of the human's mental model, which may then be refined by the user. In this paper, we leverage both human and machine intelligence to create a HILML system to aid sighted humans create video descriptions for BVI users.

System

We built a system to assist novice human volunteers in producing video descriptions using a HILML approach. Our framework uses publicly available APIs, such as Microsoft Azure Cognitive Services [23] and IBM Watson [29] for computer vision and language. These APIs provide capabilities for scene detection, generation of phrase-level captions on individual frames, face and emotion detection, text detection, speech synthesis, and sentiment analysis. We built a model based on this extracted knowledge to generate effective descriptions of video content. These descriptions are minimal and mainly focus on major scene or condition changes, primarily changes in foreground objects, people in the scene, and text.

The following describes the workflow of the system:

Input Data: Videos for which descriptions have been requested are forwarded to the model for processing.

Scene Segmentation and Key Frame Extraction: The video is segmented into a sequence of scenes of varying time spans. Key frames are sampled to maintain the appropriate granularity of the scene for generating the description.

Generating Video Description: Sequences of key frames are processed by the model to generate descriptions that best explain the scene in the video. The description covers any text in the key frames, people with ID (to handle reappearances), gender, emotion, hair color, age, objects, and environment. People are recognizable if they are known celebrities.

Validating or Revising the Description: Through the describer interface (Figures 1 and 2), sighted users can view video scene segments, their associated descriptions, and can make their own edits and notes in the script editor. Vol-

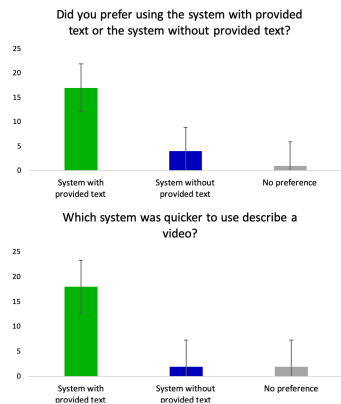


Figure 3: Describers' ratings of the HILML and Human-Only conditions. Describers showed (Top:) noticeable preference for the HILML system and (Bottom:) felt it was quicker to use.

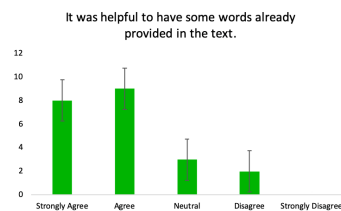


Figure 4: Describers' ratings of the helpfulness of the provided text generated by the HILML system. Most describers agreed or strongly agreed that it was helpful.

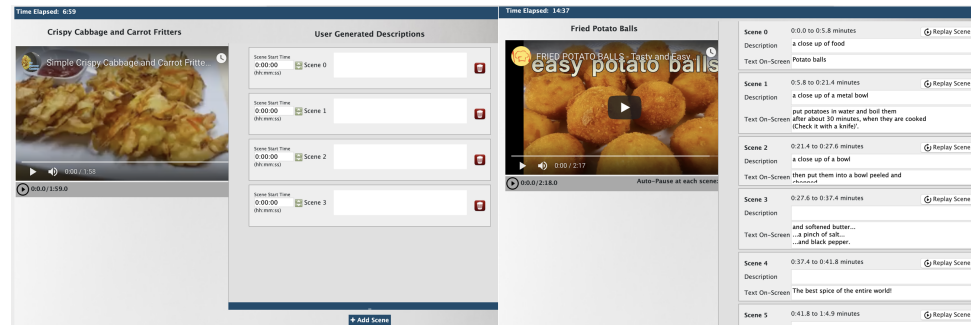


Figure 2: The Descriptor User Interface for the sighted volunteers in Experiment 1 across the Human-Only (Left Panel) and HILML (Right Panel) conditions. The video is played on the left-hand side and the edit/input interface is on the right. In the Human-Only condition, describers need to create their own scene segmentations and generate all of the descriptions from scratch. In the HILML condition, the scenes are automatically segmented for users.

unteers improve this narration by revising, merging, or completing the machine-generated descriptions. Compared to free-form descriptions that volunteers create from scratch, the proposed process assists sighted volunteers to structure and complete the descriptions. Figure 1 shows the the scene segmentation, text generation for description of images in the video, and text generation for text that appears on the screen.

Playing Video with Description: The video is played to the visually impaired or blind user with audio descriptions synchronized to the original video. Audio is generated using IBM Watson's text-to-speech API.

Experiment 1: Describing Videos

Experiment 1 investigates the use of the HILML system for video description generation for novice describers. Participants described videos in two conditions: one with the HILML system with machine learning assistance for text generation and scene segmentation, and one with no machine learning assistance (Figure 2).

Experimental Design

In Experiment 1, twenty two participants (11 male, 11 female), aged 18 to 34 (mean age of 24.0, SD of 4.45) took part in a within-subject design and will be referred to as the *describers*. All describers first took a short tutorial to familiarize themselves with the video description software. Describers were then given two videos to describe, one with and one without the HILML system. In the *HILML condition*, automated text and scene division was provided which could then be edited by the describers if they wished. In the control condition, which we refer to as the '*Human-Only condition*', describers typed out all of their video descriptions from scratch in the user interface. Figure 2 show examples of both conditions in the user interface. Both videos were 'How-To' cooking videos of around 2 minutes length each. 'How-to' videos are requested by visually impaired users on YouDescribe. The 'How-To' video style requires a high degree of quality description which includes: 1) text that is on the screen to be described if it is present for greater

Describer Quotes on HILML system from Experiment 1:

"It was super helpful. It was a lot easier to describe what I was looking at just because it already had pre-written text. It helps get it started."

"I think that one was pretty much set and it again gave us all the details we needed and if we need to fix it we could so I think it made the process a lot easier and quicker."

"I feel like it gave me some type of guideline as to what I should put, and not too many details."

"More helpful part was that it was already dividing up into the different scenes because the fact that it was automatically dividing up meant that all I have to do is just create the script."

"It was helpful for starting it. But it wasn't too helpful in being more descriptive. It would be a very simple description like 'a bowl was present.'"

"[The accuracy] was kind of neutral because I also needed to interpret what it was saying. Some of the text was helpful but the other things I deleted it completely and started from scratch."

than 1 or 2 seconds, and 2) the scene to be described every time it changes. The describers were divided into two groups: Group 1 described Video 1¹ with the HILML system and Video 2² without the HILML system. Group 2 described Video 1 without the HILML system and Video 2 with the HILML system. The order of the conditions and group allocations was alternated between participants. Both videos had music playing throughout with text on screen appearing with instructions and ingredients. At the conclusion of each video description, describers were asked to fill out a questionnaire on their describing experience including an unweighted NASA-TLX survey [11] (a subjective workload assessment tool) and were given a short interview.

Results and Discussion

Questionnaire data: Questionnaire data shows that describers preferred the HILML system and felt that it was quicker than the Human-Only condition (Figure 3) even though both systems were generally easy to use for most participants. This is interesting as the accuracy of the provided text by the HILML system was reported as being somewhat accurate, with the mode response being of 'neutral' accuracy. This suggests that even though the accuracy of the HILML system was not perfect, it was *still* helpful (Figure 4) than no machine learning assistance at all.

Time Taken to Complete Video Descriptions: We performed a t-test on the mean time (in seconds) that describers spent describing a video across the HILML and Human-Only conditions. Results showed that users described videos with the HILML condition ($\mu = 1285.41$, $\sigma = 659.51$) significantly faster than the Human-Only condition ($\mu = 1825.45$, $\sigma = 658.51$) ($t(21) = 2.83$, $p = 0.005$, $d = 0.60$). These findings correspond with the questionnaire data by describers

¹ <https://www.youtube.com/watch?v=cNj3aOTYdQQ>

² <https://www.youtube.com/watch?v=nqXz8hhAYGo>

	Human-Only	HILML	Wilcoxon Z	p	effect size
Mental demand	64.1 (20.7)	43.4 (23.9)	-2.486572	0.011110	0.530139
Physical demand	19.8 (20.5)	14.8 (17.6)	-0.813411	0.427694	0.173420
Temporal demand	40.5 (29.7)	28.0 (22.0)	-1.463076	0.148385	0.311929
Performance	31.1 (22.3)	26.8 (27.8)	-0.179176	0.866728	0.038200
Effort	60.7 (18.5)	45.7 (23.3)	-2.199999	0.026089	0.469041
Frustration	43.4 (27.3)	23.6 (22.9)	-2.548356	0.008972	0.543311

Table 1: Mean (st. dev.) NASA-TLX values (0=low, 100=high) and results from Wilcoxon Signed-ranked test. Rows in bold indicate significant results.

that the HILML system was quicker to use than the Human-Only condition (Figure 3).

Describer Workload: Results of the NASA-TLX responses (Table 1) showed that describers found the HILML system to be significantly less mentally demanding, require less effort, and be less frustrating.

Interview Data: All describers were asked the following three interview questions: 1) 'How helpful was the provided text?', 2) 'Does using the provided text make describing the videos easier?', and 3) 'Is there anything else you'd like to add about the systems you used to describe the videos?'. [Representative quotes are provided in the margin.] The two main features provided by the HILML system for describers was automatic text generation for video descriptions and scene segmentation. Sixteen out of 22 describers commented on how these features helped provide guidelines for them as novices, particularly on the level of detail required in descriptions. However, four out of 22 describers felt that the descriptions were "vague" and "simple" (although two of these describers still felt the HILML system did provide them with a starting point or a guideline). Interview data was consistent with questionnaire data on the text generation's neutral level of accuracy, yet the HILML was still helped the majority of novice describers.

Rater Quotes from Experiment 2 on HILML system and Human-Only Control Condition:

“Good cooking advice, actually describes what TO do and not just what is going on. Would say excellent.”[HILML System]

“This video is an example of the minimum info that should be included in all videos. It is not too overly descriptive with each step. It lays out what is happening and then reads the text for the ingredients.”[HILML System]

“There weren’t any long pauses in the video and it felt as if the description was done more carefully and with precision to match the pace of the original video.”[HILML System]

“Too specified, too confusing to understand” [Human-Only condition]

“The details were very vague and no text shown on the screen was even mentioned.”[Human-Only condition]

“I believe that aligning an audio track with the description is much more beneficial in order to prevent any confusion.” [Human-Only condition]

Experiment 2: Video Descriptions Rated By Blind or Visually Impaired Users

Twelve BVI participants were asked to rate the videos created by describers in Experiment 1 (akin to [15]). These participants will be referred to as *raters*. Methods reported by raters to access images were audio description (9 raters), magnification (2 raters), and tactile graphics (1 rater).

Experimental Design

All raters worked remotely and were each given half of the 44 videos described (as each of the 22 describers had produced 2 videos). Six raters were given videos produced by the first 11 describers and six raters were given videos described by the last 11 describers. Therefore, half of the videos provided to each rater were described using the HILML system and half were described by the Human-Only condition. The videos were split evenly into Video 1 and Video 2. The order of the videos provided to raters were randomized. Raters were not even aware that some videos had been described using the assistance of machine learning technology, they were simply told that the videos had been described by 11 different describers. Raters could play each video through and hear the video descriptions being read out by text-to-speech synthesis. After watching each video, raters were asked to rate the quality of description on a scale of 1 to 5 (‘Poor’ to ‘Excellent’), their understanding of the topic in the video, and any other comments.

Results and Discussion

Video Description Quality and Topic Understanding: Wilcoxon Signed-ranks tests showed that mean video description quality was rated significantly higher in videos created by the HILML system ($\mu = 3.9393$, $\sigma = 0.5276$) than the Human-Only condition ($\mu = 3.4303$, $\sigma = 0.7328$) ($Z = 2.5897$, $p = 0.0068$, $r = 0.7476$) and that mean understanding of video’s topic was rated significantly higher

in videos created by the HILML system ($\mu = 4.1720$, $\sigma = 0.6650$) than the Human-Only condition ($\mu = 3.8417$, $\sigma = 0.8026$) ($Z = 2.5001$, $p = 0.0117$, $r = 0.7217$).

Comments by Raters: The comments provided by the raters were extremely interesting [please see margin for representative quotes]. The level of detail provided in descriptions was very important to raters, with the HILML system providing more concise and succinct descriptions. Novice describers often provided too much detail in the Human-Only condition. Raters highlighted many cases in the Human-Only condition where there was a lack of reading out text on screen. The HILML system automatically generates text on screen, guiding the novice describer in that respect. Correct scene segmentation and alignment of audio and video was also very important to BVI raters with the HILML system creating a more cohesive “*flow*” for raters, as opposed to a speed that was “*rushed*” in the Human-Only condition.

Conclusion

This work demonstrates that the HILML system 1) helps novice describers generate video descriptions for BVI users significantly faster and with increased ease, and 2) generates significantly higher quality video descriptions and understanding than human-only descriptions as rated by BVI users. This provides a hugely needed low barrier to entry method for online video descriptions in order to decrease the massive digital divide between BVI and sighted users. As part of future work, we are developing ‘on-demand descriptions’ to allow BVI users to pause the video and ask questions using natural language.

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Human-in-the-Loop Machine Learning to Increase Video Accessibility for Visually Impaired and Blind Users

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ABSTRACT

Video accessibility is crucial for blind and visually impaired individuals for education, employment, and entertainment purposes. However, professional video descriptions are costly and time-consuming. Volunteer-created video descriptions could be a promising alternative, however, they can vary in quality and can be intimidating for novice describers. We developed a Human-in-the-Loop Machine Learning (HILML) approach to video description by automating video text generation and scene segmentation and allowing humans to edit the output. The HILML approach facilitates human-machine collaboration to produce high quality video descriptions while keeping a low barrier to entry for volunteer describers. Our HILML system was significantly faster and easier to use for first-time video describers compared to a human-only control condition with no machine learning assistance. The quality of the video descriptions and understanding of the topic created by the HILML system compared to the human-only condition were rated as being significantly higher by blind and visually impaired users.

Author Keywords

Video Accessibility; Video Description; Blind Users; Visually Impaired Users; Machine Learning; Human-in-the-Loop;

CCS Concepts

•Human-centered computing → Accessibility technologies; Accessibility systems and tools;

INTRODUCTION

The World Health Organization (WHO) estimates that roughly 285 million people worldwide are visually impaired, and 39 million of them are blind [9]. In the United States alone, there are 10 million people who are visually impaired, of whom 1.3 million are blind [9]. Video accessibility for blind or visually impaired users¹ becomes crucial as online videos on

¹In this paper, we use the terms blind and visually impaired users to refer to individuals who rely on audio descriptions, tactile graphics, or use magnification to typically access images.

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websites such as YouTube come to play an essential role in our daily lives. However, the vast majority of video material on the Internet is currently not accessible to these millions of visually impaired people who would significantly benefit from improved access to videos for education, employment, social, and entertainment purposes, thus creating an increasing digital divide between blind and sighted users. The United Nations Convention on the Rights of Persons with Disabilities states that persons with disabilities should be able to access services on the Internet (Article 21) and enjoy access to cultural material in accessible formats (Article 30) [49]. The Web Content Accessibility Guidelines recommends that audio descriptions be provided for all online videos [15]. Despite such international guidelines and standards, there is still a paucity of videos that are made accessible through video description for blind or visually impaired users. Video description can be defined as “Narration added to the soundtrack to describe important visual details that cannot be understood from the main soundtrack alone” [8]. Video description is particularly important as the speech communication channel is considered to be one of the most important modalities for blind and visually impaired users [23].

A significant bottleneck in video accessibility for blind and visually impaired users is the time and cost to produce video descriptions professionally. Professional video description typically has a minimum two-week turnaround time and can cost up to hundreds or thousands of dollars for a video [20], depending on its length. Alternatives to professional video description include tools such as MAGpie [1] and LiveDescribe [44], which allow anyone to describe videos. The website YouDescribe [67] is a free platform with around 2500 sighted volunteers that have described over 4000 YouTube videos. However, inexperienced describers can struggle to produce high quality descriptions, with the quality of video descriptions on YouDescribe varying significantly depending on the volunteers.

Describing videos requires expertise to clearly but *succinctly* describe scenes within limited gaps in the original soundtrack. Video description can be daunting for inexperienced volunteers who can be limited by their own vocabulary, writing skills, or lack of training. Be My Eyes [21] is an application that connects blind and visually impaired people with two million sighted volunteers for visual assistance through live and generally short video calls. Informal feedback from Be My Eyes volunteers to the authors revealed that the difficulty of the task and skills required such as language and/or writing

skills, combined with the public exposure of their work, were deterrents to them volunteering for video description. This informal feedback, the large discrepancy in volunteer numbers between YouDescribe and Be My Eyes, along with the fact that Be My Eyes is currently overwhelmed with fifteen times the number of sighted volunteers to blind or low-vision users [21], suggests that people are willing to help blind and visually impaired users, but describing videos is not something that are able to or willing to currently do easily.

In this paper, we turn to the use of Human-in-the-Loop Machine Learning (HILML) to facilitate easier video description for novice describers and to improve the quantity and quality of video accessibility for blind and visually impaired users. We built a video description interface with automated scene segmentation, text generation of images, and text generation of text on screen using machine learning. The human is then able to edit the generated output, allowing human-machine collaboration to produce high quality video descriptions while keeping a low barrier to entry for volunteer describers. We evaluated this interface on novice describers and compared it to a control condition with no machine learning assistance (Experiment 1). We then asked blind and visually impaired users to rate the quality and clarity of the video descriptions produced by the two conditions (Experiment 2). Thus, the contributions of this paper are as follows:

1. Design and develop a Human-in-the-Loop Machine Learning (HILML) system to generate video descriptions for visually impaired and blind users.
2. Demonstrate that the HILML system is faster and easier to use for novice video describers than a human only control condition.
3. Demonstrate that the quality of the video descriptions and topic understanding is significantly higher for the HILML system for blind and visually impaired raters compared to a human only control condition.

RELATED WORK

Video Description and User Interfaces

There have been a wide range of innovative works in the field of HCI and accessibility for blind and visually impaired users including visual graphics [42], data visualizations [64], maps [47, 50], programming [53], board games [16], video games [27, 56, 6], and appliance displays [24]. For the purposes of this paper, we focus on research in video description and divide the related work in this area into: 1) front-end user interface designs for increased video accessibility, and 2) tools developed for the automation of video description.

Front End User Interface Design for Video Accessibility

Various researches have investigated the front-end of the user interface for increased accessibility of videos and films for blind and visually impaired users. Chapdelaine and Gagnon [13] investigated on-demand video description with a Website platform that could provide users with extended descriptions that exceeded the natural pauses in the soundtrack by pausing the video. Results showed that selecting various video description levels was appreciated by around 92% of the users.

Chapdelaine [12] demonstrated a DVD player with video description with three extra specialized functions: contextual functions, differing quantities of audio description, and recall assistance. Results showed that 94% of blind and visually impaired users appreciated having these extra functions.

Encelle et al. [20] investigated annotation-based video enrichment made up of both speech synthesis and ‘earcons’, i.e., non-verbal audio messages to provide information to the user. Results showed that earcons could be used together with speech synthesis to enhance understanding of videos when accompanied with explanations. Encelle et al. [19] investigated the use of artificial pauses in audio described videos in order to transmit more information in the video descriptions and suggest the placing of artificial pauses right after the soundtrack’s natural gaps [19].

Tools Toward Automated Video Description

LiveDescribe was the first known computer-assisted video description [11]. It was designed for real-time descriptions of live television broadcasts. The two automated features of LiveDescribe were: 1) automatic detection of silent periods where a describer could choose to add a description, and 2) allow descriptions to go beyond the silent periods by automatically pausing the video to accommodate extended descriptions. Feedback from describers was positive but also reflected the high cognitive workload of live description [11].

The Collaborative Annotation for Video Accessibility (ACAV) project developed a prototype for an adaptive video enrichment system personalized to blind and visually impaired users based on decision trees [57]. After an initial learning phase with user feedback, the system used what it learned about the user to predict feedback using a decision-tree. The idea of personalizing video descriptions to individuals is laudable, however, as stated by the authors, the prototype needs to be evaluated with user studies to investigate the adaptive system further [57].

As part of the e-inclusion project, Gagnon et al. [26, 25] built a prototype software for professional describers that used computer vision to extract visual content from videos. They used computer vision techniques such as shot transition detection, key frame identification, face recognition, text spotting, visual motion, gait/gesture characterization, key-place identification, key-object spotting, image categorization, and also detected silent areas in the soundtrack [26, 25]. Professional describers, already have expertise and training in how to sufficiently and succinctly use language for video description. Gagnon et al. [26, 25] displayed detected features on an interactive timeline which allowed professional describers to have an overview of the frequency and duration of features in the video, as well as the silent areas. This helped them to use their expertise in *placing* their text in the correct areas. Their work differs from ours in two key ways: 1) their computer vision techniques were employed to produce specific tasks such as key-face recognition, key-object spotting etc while our approach utilizes deep learning general intelligence of video description generation, and 2) their user interface was designed specifically for professional video describers while we aim to lower the barrier to entry for novice describers, with human-machine collaboration.

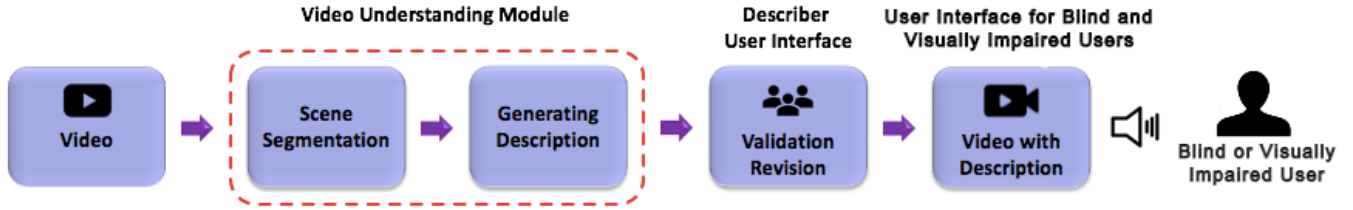


Figure 1. The workflow of the HILML system: the inputted video is segmented into scenes, automated text is generated for the images and text on screen, which can then be edited by the human. The descriptions (with the segmented timings) are synthesized to speech and outputted to play alongside the video for the blind or visually impaired user.

Kobayashi et al. created audio description synthesis by using external metadata of videos on an online platform [39, 40]. They provided a script editor to the describers, a prototype video player for the blind and visually impaired users, and metadata storage exchange between the components. However, they found there was a gap in the quality of audio descriptions produced by the software against professional narrated descriptions. Kobayashi et al. [40] tested their describer script editor, which has a visual interface to edit the video description sentences and to specify where on the video timeline to read each sentence aloud, on one novice and one professional describer. Ratings by blind and visually impaired users showed significantly lower scores for the novice describer [40]. The novice describer estimated that it would take three times longer without the script editor to produce video descriptions, but no control condition was carried out.

Machine Learning in Video Understanding

Classical video description approaches combined subject, object, and verb (SVO) detection from visual entities with template based language models to generate sentences [41]. However, the release of large datasets revealed that these methods cannot cope with the diversity in unconstrained open domain videos. The ‘Inverse Hollywood Problem’ approach tried to describe a series of actions into semantic tag summaries in order to develop a storyboard from instructional videos [10]. Many classical SVO approaches [30, 18, 7] were able to heuristically parse the videos into a series of key actions and generate a script that describes actions detected in the video. It also generated key frames depicting the detected causal events and defined the series of events into semantics representation, but the object and vocabulary/actions were very limited [3].

Classical approaches and other advancement such as statistical methods were soon replaced with deep learning, the current state of the art in video description. In particular, Convolutional Neural Networks (CNNs) [5] are the state of the art for modeling visual data recognition [5, 59, 61] and Long Short-Term Memory (LSTMs) [32] are now dominating the area of sequence modeling such as NLP [14, 29, 60]. The deep learning approaches to video description are also usually divided into two sequential stages, namely, visual content extraction and text generation. Visual features represented by fixed or dynamic real-valued vectors are produced instead. This is often referred to as the video encoding stage. CNN, Recurrent Neural Network (RNN), or LSTM are used in this encoding stage to learn these visual features, that are then

used in the second stage for text generation, also known as the decoding stage. For decoding, different flavours of RNNs are used, such as deep RNN, Bi-directional RNN, LSTM, Gated Recurrent Units (GRU), Attention or Transformer [63]. The resulting description can be a single sentence or multiple sentences. Roemmele et. al used a recurrent-neural-network architecture to generate stories in a sequence-to-sequence manner [55]. Martin et. al proposed an event representation for neural-network-based story generation [45]. Fan et al. created a hierarchical model that automatically generates stories conditioning on the writing prompts [22]. Huang et. al introduced the visual storytelling task, in which the trained model takes a sequence of photos as input and generates a short story that narrates this photo sequence [36].

Although the machine-generated descriptions are not currently as good as human-generated ones [34], the latest research described above shows potential for supporting sighted volunteers in generating video descriptions for blind and visually impaired users. Our approach assists sighted users to create video descriptions by combining automation using deep learning with human input and editing. Such an approach can be considered Human-in-the-Loop Machine Learning, which we examine next.

Human-in-the-Loop Machine Learning

Human-in-the-Loop Machine Learning (HILML) has been defined as human and machine learning (ML) processes interacting to solve one or more of the following: 1) Making ML more accurate, 2) Getting ML to the desired accuracy faster, 3) Making humans more accurate, and 4) Making humans more efficient [48]. In this paper, we aim to use HILML to make the describers more accurate and efficient, however, in the long-term, the data collected can ML to become more accurate as well. HILML is an important field to both the HCI and ML communities and can create important collaborations. For example, [48] points out that when machine translation was hitting difficulties in making human translation faster and more accurate, it was an award winning paper in human-computer interaction by Green et al. [58] that created a breakthrough with human post-editing for language translation.

HILML has been used in fields such as assistive robotics [28], data analytics [58], systems [33, 66], and audio labeling [37]. In the field of image and video classification, Wang et al. [65] used HILML to create a semi-automatic method to segment foreground moving objects in surveillance videos. They combined human outlining of a small number of moving objects

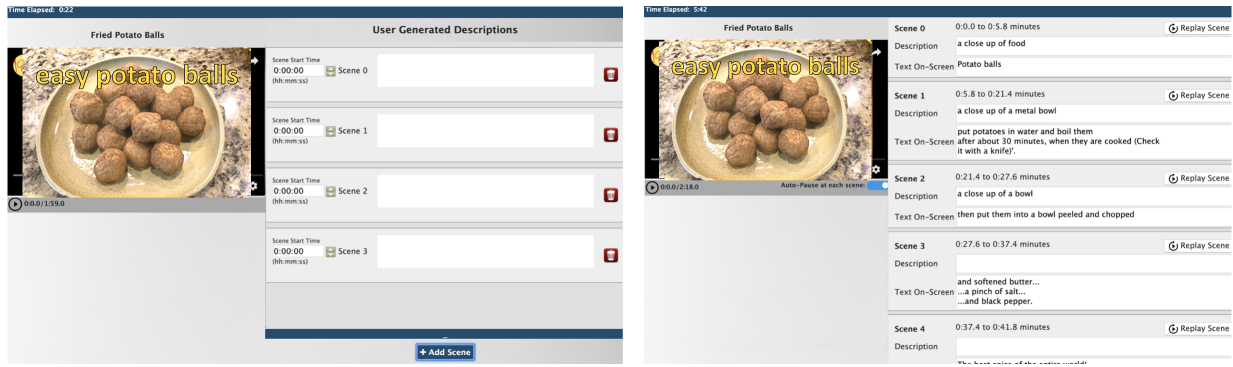


Figure 2. The Describer User Interface for the sighted volunteers in Experiment 1 across the Human-Only (Left Panel) and HILML (Right Panel) conditions. The video is played on the left-hand side and the edit/input interface is on the right. In the Human-Only condition, describers need to create their own scene segmentations and generate all of the descriptions from scratch. In the HILML condition, the scenes are automatically segmented for users. Automated description from images in the video and text on screen is generated for users in script editors, allowing users to edit the automated output. Figure 3 shows a close-up of this section.

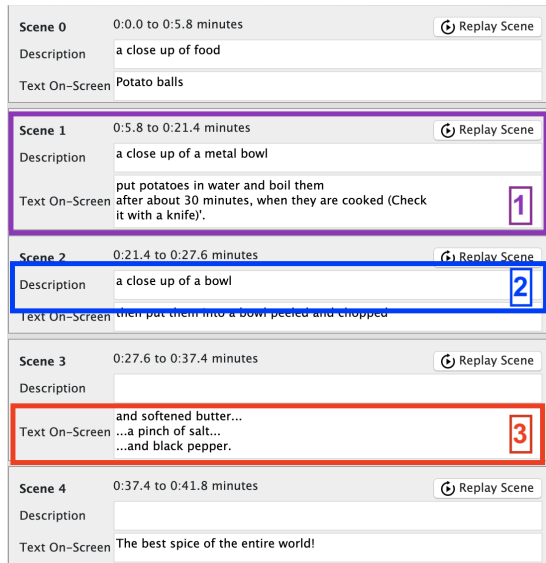


Figure 3. Close-up of the Describer User Interface and script editor of the HILML system displaying: 1) automated scene segmentation, 2) text generation for the description of images in the video, and 3) text generation for text that appears on the screen. This automated text can then be edited by the describer (the text shown here is the automated output with no human edits).

with a convolutional neural network to reach similar levels of accuracy as a human with less manual work than human-only analysis [65]. Pirrung et al. [52] created a HILML image organization web application where humans can organize their images and the machine learns from these groupings. In return, the machine can reposition images or regroup them to reflect its assessment of the human’s mental model, which may then be refined by the user.

In this paper, we leverage both human and machine intelligence to create a HILML system to aid sighted humans create video descriptions for blind or visually impaired users. We now describe the HILML system we created for video description generation.

SYSTEM

We built a system to assist human volunteers in producing video descriptions using a HILML approach. Video description including YouDescribe uses either in-line description or extended-description to add descriptions over existing video. In-line description is inserted to the original video if audio is quiet and is preferred. Extended-description is inserted by stopping video flow when the original video audio track is too busy with dialog or important sound effect. While both descriptions are helpful, descriptions are desired to be succinct and minimal while delivering important information of the scenes. With these required features, we have designed the system as below.

The following describes the workflow (Figure 1) of the developed system:

1. **Input Data:** Videos for which descriptions have been requested are forwarded to the model for further processing.
2. **Scene Segmentation and Key Frame Extraction** [Video Understanding Module]: The video is segmented into a sequence of scenes of varying time spans, and key frames are sampled to maintain the appropriate granularity of the scene for generating the description. A new scene is created when either 1) the background changes, or 2) a new character in the foreground enters or exits the scene. A new description is generated and attached to the scene. We merge any scenes with continuous dialog as the description is read out using text-to-speech technology. Describers can also merge scenes if they wish. Information extracted from the video is carried out by Microsoft Azure Video Indexer API [46]. Text scripts with the speaker identity and timestamp tags are provided by ListenByCode [43].
3. **Generating Video Description** [Video Understanding Module]: Sequences of key frames are processed by the model to generate descriptions that best explain the scene in the video. The description covers any text in the key frames, people with ID (to handle reappearances), gender, emotion, hair color, age, objects, and environment. People are recognizable if they are known celebrities. We utilize Pythia Deep Learning image description generation model [54] for each key frame image

as there are only a few key frames per scene. We used BLEU [51] scores to drop similar sentences and combined sentences to build multi sentence baseline text. Some type of text on screen is very important and can often be forgotten by volunteer describers while other types of text appear on the screen such as street signs that should not be read out. We used a threshold for time spent on screen to determine whether the text should be described.

4. Validating or Revising the Description [Describer User Interface]: Through the describer interface (Figures 2 and 3), sighted users can view video scene segments, their associated descriptions, and can make their own edits and notes in the script editor. Volunteers improve this narration by revising, merging, or completing the machine-generated descriptions. Compared to free-form descriptions that volunteers create from scratch, the proposed process assists sighted volunteers to structure and complete the descriptions. Figure 3 shows the the scene segmentation, text generation for description of images in the video, and text generation for text that appears on the screen. When the video plays, each corresponding scene gets highlighted and automatically stopped at the end of each scene to assist sighted volunteers. Volunteers can replay the working scene with a button click.

5. Playing Video with Description [User Interface for Blind and Visually Impaired Users]: The video is played to the visually impaired or blind user with audio descriptions synchronized to the original video. Audio is generated using IBM Watson’s text-to-speech API. Synthesized video descriptions have been shown to be acceptable to blind and visually impaired users [13, 40] and are used by professional video description production companies, such as 3play Media [2] and Automatic Sync Technologies [62]. ListenbyCode [43] to processes the audio track to return the script of the dialog, timestamp at each dialog, and the corresponding speakers. This information determines the scene segmentations and also detects the period where the text-to-speech converted descriptions can be inserted. If a suitable period is not found due to busy dialog, then the speech will be inserted as "extended description" that pauses the video.

EXPERIMENT 1: DESCRIBING VIDEOS

Experiment 1 investigates the use of the HILML system for video description generation for novice describers. Participants described videos in two conditions: one with the HILML system with machine learning assistance for text generation and scene segmentation, and one with no machine learning assistance (Figure 2).

Experimental Design

In Experiment 1, twenty two participants (11 male, 11 female), aged 18 to 34 (mean age of 24.0, SD of 4.45) took part in a within-subject design and will be referred to as the *describers*. All describers first took a short tutorial to familiarize themselves with the video description software. Describers were then given two videos to describe, one with and one without the HILML system. In the *HILML condition*, automated text and scene division was provided which could then be edited by the describers if they wished. In the control condition,

which we refer to as the ‘*Human-Only*’ condition, describers typed out all of their video descriptions from scratch in the user interface. Figure 2 show examples of both conditions in the user interface.

Both videos were ‘How-To’ cooking videos of around 2 minutes length each. ‘How-to’ videos are requested by visually impaired users on YouDescribe. The ‘How-To’ video style requires a high degree of quality description which includes: 1) text that is on the screen to be described if it is present for greater than 1 or 2 seconds, and 2) the scene to be described every time it changes. The describers were divided into two groups: Group 1 described Video 1² with the HILML system and Video 2³ without the HILML system. Group 2 described Video 1 without the HILML system and Video 2 with the HILML system. The order of the conditions and group allocations was alternated between participants. Both videos had music playing throughout with text on screen appearing with instructions and ingredients. At the conclusion of each video description, describers were asked to fill out a questionnaire on their describing experience including an unweighted NASA-TLX survey [31] (a subjective workload assessment tool) and were given a short interview.

Results

Questionnaire Data

Questionnaire data shows that describers preferred the HILML system and felt that it was quicker than the Human-Only condition (Figure 4) even though both systems were generally easy to use for most participants (Figure 5). This is interesting as the accuracy of the provided text by the HILML system was reported as being somewhat accurate, with the mode response being ‘neutral’ (Figure 7). This suggests that even though the accuracy of the HILML system was not perfect, it was *still* helpful (as shown in Figure 6) than no machine learning assistance at all.

Time Taken to Complete Video Descriptions

We performed a t-test on the mean time (in seconds) that describers spent describing a video across the HILML and Human-Only conditions. Results showed that users described videos with the HILML condition ($\mu = 1285.41$, $\sigma = 659.51$) significantly faster than the Human-Only condition ($\mu = 1825.45$, $\sigma = 658.51$) ($t(21) = 2.83$, $p = 0.005$, $d = 0.60$). Figure 8 shows the time taken between the two conditions. These findings correspond with the questionnaire data by describers that the HILML system was quicker to use than the Human-Only condition (Figure 4).

Describer Workload

Results of the NASA-TLX responses (Table 1) showed that describers found the HILML system to be significantly less mentally demanding, require less effort, and be less frustrating. This is consistent with the subjective questionnaire data that describers found the HILML system to be easier to use, less frustrating, and more enjoyable than the Human-Only condition (Figure 5).

²<https://www.youtube.com/watch?v=cNj3aOTYdQQ>

³<https://www.youtube.com/watch?v=nqXz8hhAYGo>

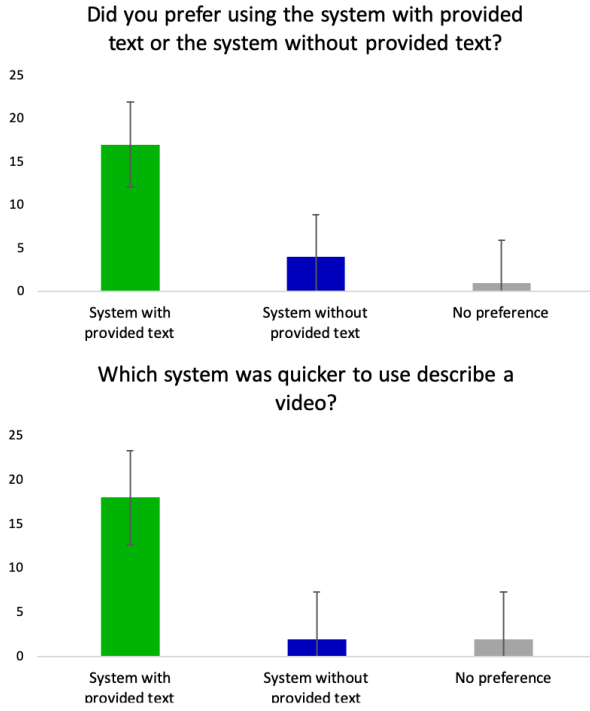


Figure 4. Describers' ratings of the HILML and Human-Only conditions. Describers showed (Top:) noticeable preference for the HILML system and (Bottom:) felt it was quicker to use.

	Human-Only	HILML	Wilcoxon Z	p	effect size
Mental demand	64.1 (20.7)	43.4 (23.9)	-2.486572	0.011110	0.530139
Physical demand	19.8 (20.5)	14.8 (17.6)	-0.813411	0.427694	0.173420
Temporal demand	40.5 (29.7)	28.0 (22.0)	-1.463076	0.148385	0.311929
Performance	31.1 (22.3)	26.8 (27.8)	-0.179176	0.866728	0.038200
Effort	60.7 (18.5)	45.7 (23.3)	-2.199999	0.026089	0.469041
Frustration	43.4 (27.3)	23.6 (22.9)	-2.548356	0.008972	0.543311

Table 1. Mean (st. dev.) NASA-TLX values (0=low, 100=high) and results from Wilcoxon Signed-ranked test. Rows in bold indicate significant results.

Interview Data

All describers were asked the following three interview questions: 1) 'How helpful was the provided text?', 2) 'Does using the provided text make describing the videos easier?', and 3) 'Is there anything else you'd like to add about the systems you used to describe the videos?'.

Sixteen out of 22 describers felt that the HILML system was helpful or easier to use because it provided guidelines:

"I think the provided text was a good set-up because it allows us to get a better idea of what the plot is versus misinterpreting it, and I think that is beneficial if you give it to anyone, everyone is going to tell you something different and this give you a better idea of where to start from."

"I think it was very helpful because it helped me to a guideline of what is in the video."

"[The provided text] provided a lot more help than just going in blank without some kind of subtext... it shows the bigger picture of what you are going for versus again describing something that may not be as relevant allowed me to save a lot more time on that."

"It was super helpful. It was a lot easier to describe what I was looking at just because it already had pre-written text. It helps get it started."

The guidelines provided by HILML text automation seemed to be particularly important for the level of detail required, with comments such as:

"I feel like it gave me some type of guideline as to what I should put, and not too many details."

"it was helpful even though I ended up using my own text. I feel like it gave me some type of guideline as to what I should put, and not too many details."

"I think that one was pretty much set and it again gave us all the details we needed and if we need to fix it we could so I think it made the process a lot easier and quicker."

"The provided text reduced the anxiety of doing it right or wrong, and to me it helped me stay focused on the task that is asked for me to do."

The HILML system also seemed to provide a guideline for scene segmentation, with four out of 22 describers making comments such as:

"More helpful part was that it was already booking up into the different scenes because the fact that it was automatically booking up meant that all I have to do is just create the script."

"Like with the second video [Human-Only], it made it a lot harder when I had to find out which time it started, which time it ended for each scene."

Four out of 22 describers felt that the descriptions were "vague" and "simple" (although two of these describers still felt the HILML system did provided them a starting point or a guideline) and made comments such as:

"It just wasn't that descriptive. It was kind of vague."

"It was helpful for starting it. But it wasn't too helpful in being more descriptive. It would be a very simple description like 'a bowl was present'."

Describer interview data was consistent with questionnaire data on the neutral level of accuracy of the HILML text generation (Figure 7) with comments such as:

"I didn't think so. I didn't think it was very accurate sometimes, and if it was just something really simple and then I just thought I could describe it better."

"It was kind of neutral because I also needed to interpret what it was saying. Some of the text was helpful but the other things I deleted it completely and start from scratch."

We discuss the implications of these findings, along with those from Experiment 2, in the overall Discussion section.

EXPERIMENT 2: VIDEO DESCRIPTIONS RATED BY BLIND AND VISUALLY IMPAIRED USERS

In Experiment 2, twelve blind and visually impaired participants were asked to rate the videos created by the describers in Experiment 1 (akin to [40]). These participants will be referred to as the *raters*. The typical method reported by raters

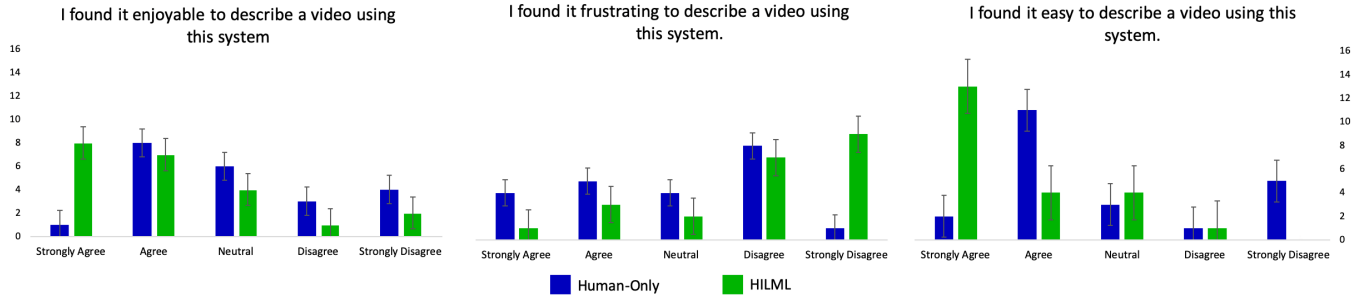


Figure 5. Describers' ratings of the HILML and Human-Only conditions for Left: enjoyment of system, Center: frustration caused by system, and Right: ease of use of system.

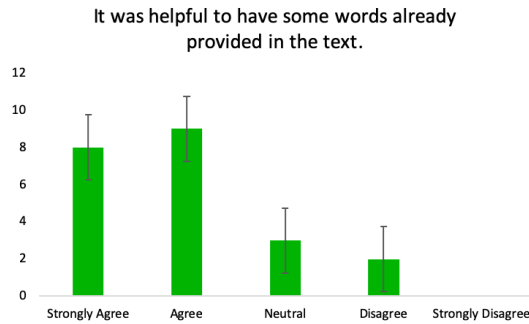


Figure 6. Describers' ratings of the helpfulness of the provided text generated by the HILML system. Most describers agreed or strongly agreed that it was helpful.

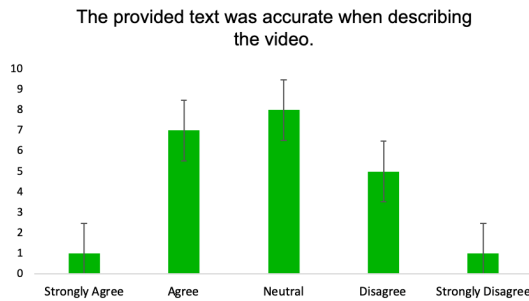


Figure 7. Describers' ratings of the accuracy of the provided text generated by the HILML system. The most highly rated level was 'neutral'.

to access images were audio description (9 participants), magnification (2 participants), and tactile graphics (1 participant). The frequency of general use of video descriptions, were daily or almost daily (5 participants), several times a week (2 participants), a few times a month (3 participants), a few times a year (1 participant), and never (1 participant).

Experimental Design

All raters worked remotely and were each given half of the 44 videos described (as each of the 22 describers had produced 2 videos). Six raters were given videos produced by the first 11 describers and six raters were given videos described by the last 11 describers. Therefore, half of the videos provided to each rater were described using the HILML system and half were described by the Human-Only condition. The videos

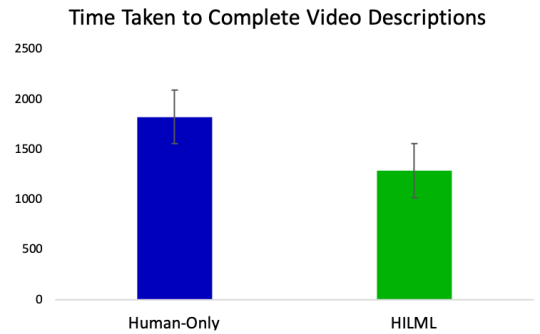


Figure 8. Mean and standard error of the time taken (in seconds) to complete the video descriptions between the HILML (green) and Human-Only (blue) conditions. The mean time taken was significantly less in the HILML condition than the Human-Only condition ($p < 0.01$).

were split evenly into Video 1 and Video 2. The order of the videos provided to raters were randomized. Raters did not know which videos had been produced by the HILML system or by the Human-Only condition. They were not even aware that some videos had been described using the assistance of machine learning technology, they were simply told that the videos had been described by 11 different describers. They were not told which pairs of videos were described by the same describer.

Raters could play each video through and hear the video descriptions being read out by text-to-speech synthesis. After watching each video, raters were asked to rate the quality of description on a scale of 1 to 5 ('Poor' to 'Excellent'), their understanding of the topic in the video, and any other comments they had about the video.

Results

Video Description Quality and Topic Understanding

Wilcoxon Signed-ranks tests showed that mean video description quality was rated significantly higher in videos created by the HILML system ($\mu = 3.9394$, $\sigma = 0.5276$) than the Human-Only condition ($\mu = 3.4303$, $\sigma = 0.7328$) ($Z = 2.5897$, $p = 0.0068$, $r = 0.7476$) and that mean understanding of video's topic was rated significantly higher in videos created by the HILML system ($\mu = 4.1720$, $\sigma = 0.6650$) than the Human-Only condition ($\mu = 3.8417$, $\sigma = 0.8026$) ($Z = 2.5001$, $p = 0.0117$, $r = 0.7217$) (Figure 9).

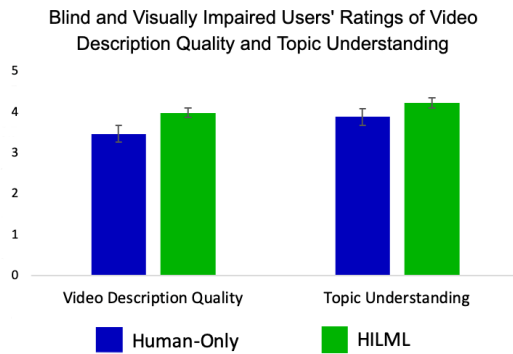


Figure 9. Mean and standard error of ratings (Likert Scale 1 (Poor) - 5 (Excellent)) made by blind and visually impaired users on the quality of video descriptions and understanding of topic in videos described by the HILML (green) and Human-Only (blue) conditions. Both video description quality ($p < 0.01$) and understanding of topic ($p = 0.01$) were rated as significantly higher in the HILML conditions.

Comments By Raters

The comments provided by the raters were extremely interesting and revealing. Out of the 12 raters, 6 of them used the word “favorite”, “excellent”, or “best” in their comments of one video. Five out of the 6 were for the HILML system. Examples of such sentences include:

“Good cooking advice, actually describes what TO do and not just what is going on. Would say excellent.” [HILML System]

“Excellent! Simple, tactile. All around easy to understand! Best so far.” [HILML System]

“This video is probably the best described, gave info on how to cut the vegetables and exact measurements.” [Human-Only condition]

The level of detail provided in descriptions was very important to raters. Novice describers are known to provide too much detail in video descriptions. Raters made comments such as: *“Too much unnecessary detail. I don’t really care that the knife is orange, but it would be helpful to know what type of knife was being used: paring knife, etc.”* [HILML system]

“Too wordy and algorithmic! Would say it is horrible. The time stamp helps NO ONE and distracts from purpose of the video. Fair.” [Human-Only condition]

“Descriptive info such as colors and sizes of items are great for videos in which that info is required. Items such as tech where people may want to know the size of things or color of lights on tech. Fashion videos where blind people may also want to know such things. But, for info videos on cooking I think it is a bit too much.” [HILML system]

“Clear and specific.” [HILML system]

“It was not necessary to say [where] the words or items were on the screen and that the words faded in or out.” [Human-Only condition]

Another category highlighted by raters were the provision of scene identification or shifts by videos that were created

using the HILML system. Closely related to this was the alignment of the audio and video, which the automated scene segmentations provided help with.

“Weird transitions, bad descriptions of bowls confused me. Lack of verbal instruction is an issue. Bad.” [Human-Only condition]

“Once again, the pace of the description does not match the pace of the video. The description went faster than the video thus leaving a huge pause in which nothing was being described only music was playing. I have a feeling the video was still going. I think the description was done very poorly.” [Human-Only condition]

Examples of positive comments by videos created by the HILML system include:

“There weren’t any long pauses in the video and it felt as if the description was done more carefully and with precision to match the pace of the original video.” [HILML system]

“All the descriptions were there. The narrator announced scene shifts as well.” [HILML system]

The reading out of text on screen was very important to raters and was highlighted in a number of comments. The HILML system automatically generates text on screen for the human, therefore the lack of text on screen transcription was most common for video descriptions created by the Human-Only condition. Examples include comments such as:

“There were far too many gaps in this description and why tell me there is instruction on the screen but not read it that does not work at all.” [Human-Only]

“No text was read and there were a large amount of grammatical errors.” [Human-Only]

“The description of this video was done very poorly. The details were very vague and no text shown on the screen was even mentioned.” [Human-Only]

“On-screen text is critical, this is a strong video description.” [HILML system]

“This first video was done very well. The person describing gave plenty of details to be able to understand what was happening. I really liked that he was reading the text that appeared on screen and read it alongside the descriptions.” [HILML system]

“I think over all the audio description was done well. It just seemed as if the description was rushed. I did like that the text that was shown on the screen was read.” [HILML system]

Many raters commented on the text-to-speech synthesis of the video descriptions across both conditions. Traditionally, in the literature, synthesized video descriptions have been shown to be acceptable to blind and visually impaired users [13, 40]. However, the synthesized voice, while acceptable created mispronunciations of words such “bowl”:

“Using an actual human as the describer would be better only because this speech synthesizer mispronounces the word ‘bowl,’ which may confuse some people.” [HILML system]

Even worse, the synthesized voice meant that there were spelling mistakes read out (such as “subscribe”) and grammatical errors which could have been corrected by a human if one was to read out the text:

“This description is a bit hard to follow. Some of the sentences don’t have a period or other punctuation to close out, so they run together with other sentences, making the description a bit too much to understand throughout.” [HILML system]

DISCUSSION

Results from Experiment 1 showed that novice describers found the HILML system to be easier and quicker to use even though it was only somewhat accurate. The time taken to describe videos was significantly less for the HILML system and describers reported the HILML system to be less mentally demanding, requiring less effort, and to be less frustrating using NASA-TLX measures. Interview data showed that describers found the HILML system to be useful in providing guidelines for the level of description detail and scene segmentation. Experiment 2 showed that blind and visually impaired users found the videos that were described by the HILML system to have a significantly higher quality of video description and understanding of the topic than the Human-Only control condition. The raters’ comments reveal a multi-faceted picture as to why this could be.

We discuss the important factors highlighted by describers (Experiment 1) and blind and visually impaired raters (Experiment 2), and discuss the implications for the HILML system.

Level of Description Detail

One of the main features that the HILML system provided was automatic text generation for video descriptions. We look into this further by examining the level of detail provided in the video descriptions, which was an important factor for both describers and raters. One of the biggest issues encountered by novice describers, and for the blind and visually impaired users who watch their video descriptions, is that novice describers have no training or knowledge of the level of detail that is required for video descriptions. Video descriptions need to be succinct, clear, and to the point, of course, this can vary based on the type of the video that is being described and can vary with individual preferences across blind and visually impaired users. While too little detail can be provided, usually, novice describers can often err on the side of more is better. This can be illustrated by rater comments such as: *“Too specified, too confusing to understand”* [Human-Only condition] or *“Unnecessary descriptions may confuse viewers ‘cut pieces into even smaller pieces’ does not help. Bad.”* [Human-Only condition].

The quality of the video description was rated significantly higher when created by the HILML system compared to the Human-Only control condition, and this is backed-up by comments from raters such as *“This video is an example of the minimum info that should be included in all videos. It is not too overly descriptive with each step. It lays out what is happening and then reads the text for the ingredients.”* [HILML System]. That is not too say that the HILML system level of detail was perfect, but it does suggest that the HILML system

is providing some type of guidance on the level of detail for describers through text automation. This is consistent with feedback from describers on the HILML system such as *“it helped me to a guideline of what is in the video”* and *“I feel like it gave me some type of guideline as to what I should put, and not too many details.”*

On the topic of detail, the HILML system captured the subtitles/text on screen, and it is extremely worthwhile to note that some raters commented on the importance of this. For example one rater watching a video created by a Human-Only condition commented: *“The details were very vague and no text shown on the screen was even mentioned.”*

Scene Segmentation

Another important feature of the HILML system was scene segmentation of the videos, providing novice describers with a framework of where to put their video descriptions. This was reflected in describer feedback such as: *“More helpful part was that [HILML condition] was already booking up into the different scenes”* or *“Like with the second video [Human-Only condition], it made it a lot harder when I had to find out which time it started, which time it ended for each scene.”* Such scene segmentation was also very important to blind and visually impaired raters as can be demonstrated by comments such as: *“I believe that aligning an audio track with the description is much more beneficial in order to prevent any confusion.”* [Human-Only condition]. The HILML system also seemed to create a more cohesive “flow” for raters, as opposed to a speed that was “rushed” in the control condition.

Such aligning of the audio and video tracks and correct division of video segments, is crucial to the understanding of the topic and may well be part of the reason why the HILML system created videos that were rated with significantly higher understanding of the topic than the Human-Only condition.

Text to Speech

The videos were all played back to blind and visually impaired raters using synthesized voices with text-to-speech technology. Several raters commented that while speech synthesis is generally acceptable, there were mispronunciation of words such as “bowl”. One rater commented that: *“The voice didn’t seem very natural and sentences that were being spoken sounded as if they were put together with no pauses in between.”* This suggests that the synthesized voice could take away from the clarity of the descriptions.

The raters also highlighted another problem with the text-to-speech technology which was that there was no human reading out the text to check for spelling mistakes and grammatical errors. There were several comments by raters across both videos created by both the HILML and Human-Only conditions of such errors that were being read out incorrectly by the text-to-speech converter. Of course, the describer of the ‘Human-in-Loop’ part of the system should have been checking for these but volunteer describers either created errors or let text with spelling mistakes and grammatical errors pass through to the blind and visually impaired users. For example, one rater said: *“I believe that using an actual person as a*

describer would be more beneficial so as to prevent misunderstandings brought on by spelling errors.” An additional note on this by a blind user who gave informal feedback to the authors was that blind and visually impaired individuals themselves may wish to lend their own voices to pre-generated text for video descriptions.

Increased Speed and Ease of Use for Novice Describers

Describers gave feedback that the HILML system was easier and quicker to use than the Human-Only condition. This was backed up by statistical analysis on time taken to describe videos and NASA-TLX subjective workload assessments. These results are very important for the blind and visually impaired community just for the simple fact that systems that automate text generation and scene segmentation, even if they need human input and editing, still make the job much easier for volunteer describers. If such a system could lower the barrier to entry for volunteer describers, this could start increasing the number of video descriptions online and start decreasing the huge digital divide and accessibility between blind and sighted individuals.

We were hoping that the HILML system would at least lower the barrier to entry for describers, that in itself, would have been progress. The fact that the HILML system also increased the video description quality is a step forward on two fronts.

Limitations

It is important to acknowledge that the videos being described in this study had music in the background and text on the screen. There was no speech or dialogue occurring during the videos. Background speech requires a more complex placement of descriptions into the video. Descriptions need to be placed where there is no background speech or sometimes the video will need to be paused for the description to be inserted. This can be challenging for novice describers, especially as they have not been trained on how much detail they should provide. We address this in our future work section with an ‘on-demand’ HILML approach.

It is also important to acknowledge that the accuracy of the text generation was rated as being mainly neutral by describers. The text-generation machine learning technology is still a work-in-progress. However, despite the neutral accuracy of the machine learning, describers still found the system easier and quicker to use than no help at all, and blind and visually impaired users still found the video descriptions to have a higher quality. This suggests that it is the bigger picture of the video with scene segmentation and the guidance on level of detail that the HILML system is providing which is what the describers and raters are finding helpful.

Finally, the Microsoft Cognitive Services is not at the point where it can generate connected sentences that tell a story. The task of multi-image cued story generation, such as the visual storytelling dataset (VIST) challenge, is to compose multiple coherent sentences from a given sequence of images [38]. These multiple coherent sentences are currently not at the level needed to replace the ‘human’ of the Human-in-Loop [35], but would improve the speed and accuracy substantially more. The current sentences still lack descriptions of small

actions or detail. We address this in our future work section with a ‘re-training’ HILML approach.

FUTURE WORK

We are currently working on improving the accuracy and quality of the machine-generated descriptions. To this end, the discrepancy between the machine-generated and revised narrations are recorded and the revised versions are being used as input to retrain and improve the accuracy of the machine learning model. We also plan to train a visual storytelling model [36] to generate a more comprehensive narrative of the scene with spatiotemporal coherency.

Through informal feedback from visually impaired and blind users, we identified that ‘on-demand descriptions’ would be helpful when the user pauses the video to ask questions using natural language. For on-demand descriptions, we will train visual question answering [4] and visual dialog [17] models to hold a meaningful dialog with blind and visually impaired users in natural, conversational language about the content of the video.

Furthermore, we plan to integrate the HILML system into the YouDescribe website. As Experiment 1 with describers showed, the HILML system was easier and quicker to use than the Human-Only condition. We will thus investigate whether our HILML system will reduce the barrier-to-entry for volunteer describers on YouDescribe and increase the number of video descriptions online.

Finally, we will compile and release a standardized, well-structured video description data set under a free, open-source license to help the machine learning and vision communities advance the state of the art in video understanding.

CONCLUSION

We designed and developed a Human-in-the-Loop Machine Learning (HILML) approach to automatically generate video descriptions and scene segmentation for online videos. Human describers could then edit the automated text generated by machine learning. Our aim was to improve video accessibility by increasing video description ease and speed for novice describers as well as increasing video description quality for visually impaired and blind users. We compared the HILML system to a Human-Only control condition with no machine learning assistance with novice describers. Results showed that the HILML system was significantly faster and easier to use, with comments from describers pointing to guidance provided by the HILML system on level of text detail and scene segmentation. Blind and visually impaired users rated the videos described the HILML and Human-Only conditions (with no knowledge of the conditions). Results showed that they rated the videos created by the HILML system as having significantly higher quality of video description as well as significantly higher understanding of the topic presented in the video than the Human-Only condition. Extensive comments provided by blind and visually impaired users indicate the importance of several factors discussed in this paper.

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